

CivicPulse — Intelligent Civic Issue Lifecycle & Risk-Aware Routing Platform

Master Implementation Blueprint · Technofora '26 CodeCraft Hackathon · SmartCity Tech Track

Team Project Plan

Generated for step-by-step AI-assisted implementation

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0. How To Use This Document (Read This First)

This document is written to be handed to an AI coding assistant **one numbered step at a time**, not all at once. It is organized so that:

1. **Sections 1-9** give the AI the *context* it needs before writing any code: the problem, the architecture, the data model, and the API contract. Paste these once at the start of your session so the AI has the full picture.
 2. **Section 10 (“Step-by-Step Build Plan”)** is the actual execution script. Each step is self-contained: it names the exact files to create/edit, states the goal in one sentence, and gives complete, runnable code. Feed the AI **one step at a time**, wait for it to finish, verify it (run it, check the file), then move to the next step. Do not skip steps — later steps assume earlier files exist.
 3. **Sections 11-20** are reference material (algorithm deep-dives, stretch features, compliance checklist, demo script). Pull these in only when the corresponding build step needs them, or when you’re polishing / preparing the pitch.
 4. Every code snippet is written to be copy-paste runnable inside the stack chosen in Section 6 (Java 17 + Spring Boot 3 + PostgreSQL + React + Leaflet + a small Python offline script). If your team’s comfort zone is different (e.g., Node/Express instead of Spring Boot), tell the AI “port this step to Node/Express” — the algorithms and API contracts in this document don’t change, only the syntax does.
 5. Treat each “STEP” as a prompt template: copy the whole step block (goal + files + code) into the AI chat verbatim, and add “implement exactly this, don’t invent extra scope” — this is what keeps a coding AI from wandering off-spec.
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1. Executive Summary

Track: SmartCity Tech — Civic Engagement & Urban Mobility Software (Problem Statement 6 of 6, Technofora ‘26 CodeCraft Hackathon, ISA Students’ Chapter, Nirma University).

One-line pitch: *An intelligent civic issue lifecycle platform that converts citizen observations into prioritized, location-aware tasks and coordinates their resolution across city authorities and field workers — powered by the same weighted road graph that also tells citizens the safest way home.*

Why this beats a plain “report an issue” app: Most SmartCity teams will build some version of *citizen uploads a photo → pin appears on a map*. That satisfies the mandatory UI requirement but has almost no algorithmic depth, which caps how well it can score on **Technical Implementation (30%)** and **Innovation & Originality (25%)** — the two largest slices of the judging rubric. CivicPulse keeps that same simple citizen-facing flow (so it’s just as easy to demo) but backs it with:

- A **real road-network graph** (pulled from OpenStreetMap for one campus/ward) instead of a bare list of complaints.
- **Duplicate/cluster detection** (geometry + time, no ML needed) so 20 reports of the same pothole become 1 tracked civic issue, not 20 tickets.
- A **defensible priority formula** (severity × cluster size × recency × structural criticality) instead of “most-voted first.”
- **Structural criticality via graph centrality** (Brandes’ betweenness centrality) — segments near schools/hospitals/main roads get flagged automatically, without anyone manually tagging them.
- A **risk-weighted shortest-path router** where “fastest route” and “safest route” are the *same* Dijkstra/A* function called with two different edge-cost functions — a one-line

difference that is very easy to demo live and very easy to defend under technical Q&A.

- An **AI classification layer** (Gemini) that only does classification/extraction — never the assignment decision — so the interesting decision logic lives in your own backend code, which is exactly what judges will ask you to walk through.

The three interfaces: Citizen App (report + track), Authority Dashboard (verify, prioritize, assign, monitor), Worker App (task list, navigate, resolve). See Section 7.

Judging-criteria alignment at a glance:

Judging Criterion	Weight	How CivicPulse addresses it
Innovation & Originality	25%	Shared graph powering both issue triage <i>and</i> routing; duplicate clustering; centrality-based criticality — none of this is “just another CRUD app.”
Technical Implementation	30%	Real algorithms (Haversine clustering, Dijkstra/A* with swappable cost functions, Brandes’ centrality, weighted priority scoring) that a team member can explain line-by-line.
Impact & Viability	20%	Solves the real crowdsourcing-noise problem (duplicate reports), gives authorities SLA/accountability visibility, and the routing feature has obvious value beyond the hackathon (monsoon flood-risk routing, etc.).
Design & UX	15%	Three clean, role-specific interfaces; color-coded map pins; a live fastest-vs-safest toggle that’s satisfying to demo.
Live Pitch & Q&A	10%	Because the core logic is transparent (formulas, not a black-box model), every team member can explain and defend their piece — which is also mandatory under the Technical Rules (Rule 17-18).

2. Problem Statement & Rulebook Alignment

2.1 Track requirement (verbatim intent from the rulebook)

“Urban communities thrive when public infrastructure is reliable, streets are safe, and transportation flows efficiently, but cities often struggle to identify and address neighborhood problems in real time... This track focuses on building citizen-centered software, community dashboards, and mobile applications that improve urban living and civic participation.”

CivicPulse directly targets: crowdsourcing neighborhood feedback (issue reporting), enhancing safety (risk-weighted routing), and making communal services more responsive and transparent (authority dashboard + SLA tracking).

2.2 Mandatory Project Requirement — Functional UI

The rulebook is explicit: *“Pure back-end scripts, terminal-only commands, raw database pipelines, or headless code repositories will not qualify on their own.”* Every algorithm in this plan (dedupe, priority scoring, graph routing, centrality) is wrapped behind a real screen a non-technical person can use — see Section 7. Do not, under any circumstances, ship the graph/algorithm layer without the citizen web app and admin dashboard in front of it — that would fail the mandatory requirement regardless of backend sophistication.

2.3 Technical Rules compliance this plan is designed around

These are non-negotiable per the Technical Rules PDF and are woven into the step-by-step plan (Section 10) and the checklist (Section 19):

- One public GitHub repo, organizers added as collaborators.
- Commit **every 1-2 hours**, meaningful messages, no single end-of-hackathon dump (Rule 3-4). Section 10 is deliberately broken into small steps so each step becomes 1-2 commits.
- Every team member makes **at least 2 real commits** (Rule 2) — Section 18 includes a suggested task split by member so this happens naturally.
- README.md must cover: problem statement, proposed solution, features, tech stack, architecture, APIs, database, setup instructions, how to run (Rule 6) — a ready-to-fill template is in Section 16.
- No API keys/secrets in source or GitHub; use .env + .env.example (Rules 9-10) — see Section 16 for the exact .env.example.
- Disclose AI-generated code/content in the PPT (Rule 11) — Section 17 includes suggested disclosure wording.
- Working prototype/MVP required, core pitched features must be demonstrable (Rule 7) — this is why Section 3 has a hard “Must-Ship” cut line before any stretch feature.
- Final demo video ≤ 5 minutes (Rulebook, Final Submission #2) — Section 17 has a shot list sized to fit that limit.

3. Scope Decision: Must-Ship vs Stretch

Given fixed hackathon hours, everything below is split into **Must-Ship** (this alone satisfies the mandatory UI rule and is a fully demoable MVP) and **Stretch**, ordered by payoff-per-hour. Section 10's step numbering follows this exact order — stop after the Must-Ship steps if you're short on time, and each stretch step is independent enough to skip without breaking anything already built.

3.1 Must-Ship (build in this order)

1. **Citizen Web App** — photo upload + auto-geotag report form, category picker, Leaflet map with color-coded pins by priority.
2. **Backend core** (Spring Boot) — ingest endpoint, Haversine-based duplicate clustering, priority scoring, REST API.
3. **Admin/Authority Dashboard** — prioritized list, map view, “days open” indicator, verify/reassign controls.
4. **Road graph + point-to-point router** over one real, small area (pulled via the free Overpass API — no key required) with a working Dijkstra/A* shortest path.
5. **Worker App (minimal)** — task list sorted by priority + distance, “navigate” button (deep-link to Google/OpenStreetMap directions), mark-resolved flow.

3.2 Stretch (ranked by payoff-per-hour, pick from the top down)

1. **Fastest vs Safest route toggle** — cheap once the graph exists (Step 4 above); very strong live-demo moment because it's a visible, tappable UI change with an obviously different route drawn on the map.
2. **Worker auto-assignment algorithm** (domain + priority + distance + current workload) instead of manual assignment only.
3. **City Health Score per ward** — one aggregate number + map choropleth; very visual, cheap to compute once issues have categories and wards.
4. **Reporter trust score** — dampens spam reports; even a simple weighted average is enough to build, and you can describe a PageRank-style extension in Q&A even if not fully implemented.
5. **SLA / accountability view** (“this pothole has been open 11 days, past the 7-day target for its category”) — strengthens the Impact/Viability score.
6. **Heatmap layer** (Leaflet.heat) over reported issues by category — visually impressive, ~30 minutes of work once data exists.
7. **AI-based issue classification** (Gemini vision + text) replacing/augmenting the manual category picker — remember to disclose in the PPT per Rule 11, and keep a keyword-based fallback for demo reliability if the API is flaky.
8. **AI voice complaint intake** (speech → text → structured complaint) — accessibility angle for elderly/low-literacy users.
9. **Before/after resolution verification** (compare before/after photos, flag confidence) — impressive in a live demo, but highest implementation risk; do this last.
10. **Predictive flood/hazard alerts** (weather API × complaint history) — timely Gujarat-monsoon angle, reuses the same graph; good pitch material even as a partially-mocked feature.
11. **Citizen reward/leaderboard system** — nice-to-have engagement mechanic, lowest priority.

Practical note: Keep your demo area small and real (your campus or one ward) rather than “the whole city.” A live route computed over actual streets beats a bigger claim backed by fake data, and it is far easier to defend when a judge asks “is this real data?”

4. System Architecture

4.1 High-level component diagram

```
flowchart TD
    A["Citizen App (React + Leaflet)"] -->|"POST /api/reports (photo, text, GPS)"| B["Backend API (Spring Boot)"]
    B --> C["Gemini AI: classify (type, severity, department)"]
    C --> B
    B --> D["Duplicate/Cluster Engine (Haversine + time window)"]
    D --> E["PostgreSQL + PostGIS"]
    B --> F["Priority Scoring Engine"]
    F --> E
    B --> G["Road Graph Service (in-memory, precomputed centrality)"]
    G --> H["Overpass API (OSM extract, offline preprocessing)"]
    A -->|"GET /api/route?mode=fastest|safest"| G
    I["Authority Dashboard (React)"] -->|"verify / reassign / monitor"| B
    J["Worker App (React, mobile-first)"] -->|"task list, navigate, mark-resolved"| B
    B --> K["Worker Assignment Engine"]
    K --> E
    B -->|"resolution update"| A
```

4.2 Components and responsibilities

Component	Responsibility	Where in this doc
Citizen Web App	Capture photo + optional description + GPS, submit report, track status, request fastest/safest route	Section 7.1, Step 6-7, 12
Backend API	Single Spring Boot service exposing REST endpoints for reports, routing, dashboard, worker actions	Section 8-9, Step 3-5
Gemini Classification Service	Given photo + text, returns { issueType, severity, department } — classification/extraction only, never the assignment decision	Section 11.5, Step 13
Duplicate/Cluster Engine	Haversine distance + time-window clustering of incoming reports into one civic issue	Section 11.1, Step 8
Priority Scoring Engine	Deterministic formula combining severity, cluster size, recency, structural criticality	Section 11.2, Step 9
Road Graph Service	In-memory weighted graph built from OSM data; runs Dijkstra/A* on demand with a pluggable edge-cost function; centrality precomputed offline	Section 11.3-11.4, Step 10-11

Component	Responsibility	Where in this doc
Worker Assignment Engine	Scores eligible workers by domain match, distance, current workload, and priority to suggest an “optimal worker”	Section 11.6, Step 15
Authority Dashboard	Verify AI classification, adjust priority, assign/reassign workers, view map + analytics	Section 7.2, Step 14
Worker App	Prioritized task list, one-tap navigation (deep link to maps), before/after photo capture, mark-resolved	Section 7.3, Step 16
PostgreSQL (+ PostGIS optional)	Persistent store for reports, clusters, users, workers, assignments	Section 8

4.3 Why one shared graph powers two products

Both “which issue is most urgent” and “which route is safest” ultimately ask the same underlying question: *how risky/important is this road segment right now?* CivicPulse keeps a single road graph in memory where:

- **Nodes** = intersections / OSM way endpoints.
- **Edges** = road segments, each with a `baseTravelTimeSeconds`, a `centralityScore` (pre-computed once, offline), and a live `openFlaggedReports` count with severity.

The **Issue Pipeline** reads `centralityScore` to compute `criticality_multiplier` in the priority formula (Section 11.2). The **Routing Engine** reads `openFlaggedReports` to compute a live risk-adjusted edge cost (Section 11.4). Same graph, two read paths, two very different-looking product features — that’s the “one line of difference, easy to demo live” trick.

5. Core Innovation Pillars (Consolidated)

This merges both the “issue lifecycle” idea and the “weighted graph” idea into one list, each tagged Must-Ship or Stretch (matches Section 3):

1. **AI-based issue classification** (Gemini: photo + description → issue type, severity, department) — *Stretch, but high value; keyword fallback keeps demo safe.*
2. **Automatic geolocation** at capture time (lat/lng/timestamp stored with every report) — *Must-Ship.*
3. **Duplicate/cluster detection** — same location + similar issue type within a time window merges into ONE civic issue instead of N tickets — *Must-Ship.*
4. **Deterministic priority scoring** — $\text{severity} \times \text{cluster_size} \times \text{recency_decay} \times \text{criticality_multiplier}$ — *Must-Ship.*
5. **Structural criticality via graph centrality** (Brandes’ betweenness centrality over the OSM road graph, precomputed once) — *Must-Ship (precompute), feeds Item 4.*
6. **Department/authority routing** — classified issue routed to the correct authority queue — *Must-Ship.*
7. **Worker allocation** — score eligible workers by domain, priority, distance, current workload → suggest optimal worker — *Stretch (manual assignment is Must-Ship; auto-suggestion is Stretch).*
8. **Risk-weighted routing (fastest vs safest)** — same Dijkstra/A* function, two edge-cost functions — *Must-Ship the router; the toggle UI is the first Stretch item.*
9. **Navigation handoff** — deep-link the worker/citizen into Google Maps / OSM for turn-by-turn — *Must-Ship (simple deep link).*
10. **Authority dashboard analytics** — totals by status/category, map view, verify/reassign controls — *Must-Ship.*
11. **Heatmap visualization** — pothole/garbage/water hotspots on the map — *Stretch.*
12. **City Health Score** per ward, aggregating road/lighting/water/garbage/safety sub-scores — *Stretch.*
13. **Predictive alerts** — weather API × historical drainage/waterlogging complaints → flood risk banner — *Stretch, highest novelty, can be partially mocked for the demo if time runs out.*
14. **Before/after resolution verification** — compare before/after photos, output a confidence score — *Stretch, highest implementation risk, do last.*
15. **Reporter trust score / anti-spam weighting** — *Stretch.*
16. **SLA / accountability tracking** (“open 11 days, target 7”) — *Stretch.*
17. **Citizen reward system / leaderboard** — *Stretch, lowest priority.*
18. **AI Civic Copilot** (citizens ask natural-language questions like “why is my complaint delayed?”) — *Stretch, only attempt if all of the above is done — it’s a nice-to-have chat wrapper around data you already have.*
19. **AI voice complaint intake** — *Stretch, accessibility angle.*
20. **Route Safety Score for arbitrary home→destination queries** (% score with road quality / streetlight coverage / reported issues) — *Stretch, natural extension of Item 8 once routing works.*

6. Recommended Tech Stack

Layer	Choice	Why
Backend	Java 17 + Spring Boot 3 (Maven)	Strong typing catches mistakes fast under time pressure; Spring Data JPA removes boilerplate SQL; easy to explain in Q&A (Rule 17 explicitly asks about architecture/algorithms/APIs).
Database	PostgreSQL 15 (PostGIS extension optional but recommended for geo queries)	Native support for geo indexing if you enable PostGIS; otherwise plain lat/lng columns + the Haversine formula in Section 11.1 work fine without any extension.
Citizen App / Admin Dashboard / Worker App	React 18 + Vite, Leaflet.js for maps, Tailwind CSS for styling	Fast to scaffold, huge community support, Leaflet needs no API key (unlike Google Maps JS SDK), satisfies the “no committed API keys” rule trivially for the map layer.
Offline graph preprocessing	Python 3 + OSMnx + NetworkX (run once, not per-request)	OSMnx pulls real OSM road data for a bounding box via Overpass in ~2 lines; NetworkX has a ready-made Brandes’ betweenness centrality implementation — no need to hand-roll it under time pressure. Exports a small JSON graph consumed by the Java backend at startup.
AI classification (stretch)	Google Gemini API (multimodal: image + text in, structured JSON out)	Free tier available for hackathons; strong image understanding for issue photos.
Weather data (stretch)	OpenWeatherMap free tier or Open-Meteo (no key required)	Feeds the predictive-alerts stretch feature.
Auth	JWT (citizen/authority/worker roles) via Spring Security	Simple to reason about, no external dependency.
Deployment	Render / Railway (backend + Postgres), Vercel / Netlify (frontend)	All have generous free tiers suitable for a hackathon demo link (Rule 15 requires a working deployment link where applicable).

If your team is more comfortable with Node.js/Express + MongoDB or Django + PostgreSQL, every algorithm and API contract in this document ports directly — only the code snippets in Section 10 need translating. Tell the AI: “port Step N to

Node/Express” and paste the step; the JSON contracts stay identical so frontend code never needs to change.

6.1 Repository layout

```
civicpulse/
├── README.md
├── .env.example
├── .gitignore
├── backend/                                # Spring Boot service
│   ├── pom.xml
│   └── src/main/java/com/civicpulse/
│       ├── CivicPulseApplication.java
│       ├── config/
│       ├── controller/
│       ├── service/
│       ├── repository/
│       ├── model/
│       └── graph/
├── graph-preprocessing/                    # one-off Python script
│   ├── requirements.txt
│   └── build_graph.py
├── frontend-citizen/                       # React app
├── frontend-dashboard/                     # React app (or a second route in the same app)
├── frontend-worker/                       # React app (or a second route in the same app)
├── docs/
│   ├── architecture-diagram.png
│   └── api-reference.md
```

For hackathon speed, frontend-citizen, frontend-dashboard, and frontend-worker can absolutely be three role-gated routes inside **one** React app (/, /admin, /worker) instead of three separate projects — recommended unless your team specifically wants to split frontend work across three people in parallel.

7. The Three Interfaces (Screen-Level Detail)

7.1 Citizen App

Screen: Report an Issue - Camera/photo upload button (required) - Optional free-text description field (also accepts voice input in the stretch version) - Category picker: Pothole / Streetlight / Water Leakage / Garbage / Safety / Other (pre-filled automatically once Gemini classification is wired in, but always editable by the citizen) - Auto-captured GPS coordinates + timestamp (shown on a small embedded map pin for confirmation before submit) - Submit button → POST /api/reports

Screen: Map / Track - Leaflet map centered on the citizen's ward, color-coded pins: - [HIGH] High priority, [MED] Medium, [LOW] Low - Tapping a pin shows: issue type, priority, "reported N minutes ago", location, assigned worker (if any), status - "My Reports" list with live status (Reported → Verified → Assigned → In Progress → Resolved)

Screen: Route Planner (Stretch) - Two text inputs (Home / Destination) or "use current location" - Two buttons: **Fastest Route** / **Safest Route** - Map draws the selected route; a small summary line shows distance, ETA, and (for safest) how many flagged-unsafe stretches are avoided and the % detour cost — this is the single best live-demo moment in the whole project (see Section 3.2, Item 1).

7.2 Authority Dashboard

Screen: Overview - KPI cards: Total Issues, Pending, In Progress, Resolved - City map with clustered issue markers, filterable by category/status/priority - Category breakdown (Potholes / Street Lights / Water / Garbage counts)

Screen: Issue Queue - Table/list sorted by priority score (desc), each row: category, cluster size ("20 citizen reports"), priority score, age, status, assigned worker - Row actions: **Verify AI classification** (accept/correct), **Change priority**, **Assign/Reassign worker**

Screen: Analytics (Stretch) - Heatmap layer toggle - City Health Score per ward, shown as a choropleth or ranked list - Resolution-time trend chart

7.3 Worker App

Screen: Today's Tasks - List sorted by (priority desc, distance asc): [HIGH] HIGH / [MED] MEDIUM / [LOW] LOW badges, distance, ETA - Tap a task → detail view with photo, description, location, "Navigate" button (deep-links to Google Maps / OSM directions using the worker's current GPS as origin)

Screen: Resolve Task - Capture "before" photo (auto-pulled from the original report if available) alongside a new "after" photo - Notes field - "Mark Resolved" button → PATCH /api/reports/{id}/resolve → triggers citizen notification ("Your reported pothole has been resolved.")

8. Data Model

```
-- ===== USERS & ROLES =====
CREATE TABLE users (
  id          BIGSERIAL PRIMARY KEY,
  name        VARCHAR(120) NOT NULL,
  email       VARCHAR(160) UNIQUE NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  role        VARCHAR(20) NOT NULL CHECK (role IN
    ↪ ('CITIZEN', 'AUTHORITY', 'WORKER')),
  trust_score  DOUBLE PRECISION DEFAULT 1.0, -- stretch: anti-spam weighting
  reward_points INTEGER DEFAULT 0,         -- stretch: leaderboard
  created_at   TIMESTAMP DEFAULT now()
);

-- ===== WORKERS (extends users with role = WORKER) =====
CREATE TABLE workers (
  id          BIGSERIAL PRIMARY KEY,
  user_id     BIGINT NOT NULL REFERENCES users(id),
  department  VARCHAR(40) NOT NULL, -- ROAD / ELECTRICAL / WATER / SANITATION
    ↪ / SAFETY
  current_lat  DOUBLE PRECISION,
  current_lng  DOUBLE PRECISION,
  active_task_count INTEGER DEFAULT 0 -- current workload, used by assignment
    ↪ engine
);

-- ===== ISSUE CLUSTERS (the "one civic issue" that many reports merge into)
    ↪ =====
CREATE TABLE issue_clusters (
  id          BIGSERIAL PRIMARY KEY,
  issue_type  VARCHAR(40) NOT NULL, -- POTHOLE / STREETLIGHT / WATER /
    ↪ GARBAGE / SAFETY / OTHER
  department  VARCHAR(40) NOT NULL,
  severity    INTEGER NOT NULL, -- 1-10, from AI classification or
    ↪ manual override
  centroid_lat DOUBLE PRECISION NOT NULL,
  centroid_lng DOUBLE PRECISION NOT NULL,
  nearest_edge_id VARCHAR(64), -- FK-ish reference into the
    ↪ in-memory road graph
  status      VARCHAR(20) NOT NULL DEFAULT 'REPORTED',
    -- REPORTED / VERIFIED / ASSIGNED / IN_PROGRESS / RESOLVED
  priority_score DOUBLE PRECISION NOT NULL DEFAULT 0,
  report_count  INTEGER NOT NULL DEFAULT 1,
  assigned_worker_id BIGINT REFERENCES workers(id),
  first_reported_at TIMESTAMP NOT NULL DEFAULT now(),
  last_reported_at  TIMESTAMP NOT NULL DEFAULT now(),
  resolved_at      TIMESTAMP
);

-- ===== INDIVIDUAL CITIZEN REPORTS (raw submissions, many-to-one into a cluster)
    ↪ =====
CREATE TABLE reports (
  id          BIGSERIAL PRIMARY KEY,
  cluster_id  BIGINT REFERENCES issue_clusters(id),
  citizen_id  BIGINT NOT NULL REFERENCES users(id),
  photo_url   VARCHAR(500) NOT NULL,
  description  TEXT,
```

```

    lat          DOUBLE PRECISION NOT NULL,
    lng          DOUBLE PRECISION NOT NULL,
    ai_issue_type VARCHAR(40),
    ai_severity  INTEGER,
    ai_department VARCHAR(40),
    created_at   TIMESTAMP DEFAULT now()
);

-- ===== RESOLUTION EVIDENCE (stretch: before/after verification) =====
CREATE TABLE resolutions (
    id          BIGSERIAL PRIMARY KEY,
    cluster_id  BIGINT NOT NULL REFERENCES issue_clusters(id),
    worker_id   BIGINT NOT NULL REFERENCES workers(id),
    before_photo_url VARCHAR(500),
    after_photo_url VARCHAR(500) NOT NULL,
    notes       TEXT,
    ai_confidence DOUBLE PRECISION, -- stretch: AI-computed resolution
    ↪ confidence
    created_at   TIMESTAMP DEFAULT now()
);

-- Helpful indexes for the dedupe query in Section 11.1
CREATE INDEX idx_clusters_open_by_type ON issue_clusters (issue_type, status);
CREATE INDEX idx_reports_created_at ON reports (created_at);

```

Notes: - nearest_edge_id links a civic issue to the road-graph edge it sits on, which is how the routing engine (Section 11.4) picks up openFlaggedReports for that edge. - priority_score is recomputed (not just stored once) every time a new report merges into a cluster — see Step 9. - If PostGIS is enabled, replace the raw lat/lng pairs with a geography(Point, 4326) column and use ST_DWithin instead of hand-rolled Haversine for the dedupe query — functionally equivalent, just faster at scale. For a single-ward hackathon demo, hand-rolled Haversine (Section 11.1) is simpler to explain in Q&A and perfectly fast enough.

9. REST API Contract

All endpoints return JSON. Base path: /api. Auth via Authorization: Bearer <JWT> except where marked public.

Method	Endpoint	Auth	Purpose
POST	/auth/register	Public	Create a user (role: CITIZEN / WORKER / AUTHORITY)
POST	/auth/login	Public	Returns JWT
POST	/reports	CITIZEN	Submit a new report (photo, description, lat, lng) → triggers classification + dedupe + priority scoring
GET	/reports/mine	CITIZEN	List the citizen's own submitted reports with cluster status
GET	/clusters	AUTHORITY	List all issue clusters, sortable by priority/status/category, for the dashboard
GET	/clusters/{id}	AUTHORITY/CITIZEN	Full detail of one civic issue (all merged reports, current status, assigned worker)
PATCH	/clusters/{id}/verify	AUTHORITY	Accept or correct the AI classification
PATCH	/clusters/{id}/priority	AUTHORITY	Manually override the computed priority score
PATCH	/clusters/{id}/assign	AUTHORITY	Assign/reassign a worker (manual, or trigger /clusters/{id}/suggest-worker first)
GET	/clusters/{id}/suggest-worker	AUTHORITY	Runs the worker-assignment engine (Section 11.6), returns ranked candidate workers
GET	/workers/{id}/tasks	WORKER	Prioritized task list for one worker
PATCH	/clusters/{id}/resolve	WORKER	Submit before/after photos + notes, mark resolved → notifies citizen

Method	Endpoint	Auth	Purpose
GET	/route?fromLat=&fromLon=&toLat=&toLng=&mode=&cost=&avoid=&id=&id2=&id3=&id4=&id5=&id6=&id7=&id8=&id9=&id10=&id11=&id12=&id13=&id14=&id15=&id16=&id17=&id18=&id19=&id20=&id21=&id22=&id23=&id24=&id25=&id26=&id27=&id28=&id29=&id30=&id31=&id32=&id33=&id34=&id35=&id36=&id37=&id38=&id39=&id40=&id41=&id42=&id43=&id44=&id45=&id46=&id47=&id48=&id49=&id50=&id51=&id52=&id53=&id54=&id55=&id56=&id57=&id58=&id59=&id60=&id61=&id62=&id63=&id64=&id65=&id66=&id67=&id68=&id69=&id70=&id71=&id72=&id73=&id74=&id75=&id76=&id77=&id78=&id79=&id80=&id81=&id82=&id83=&id84=&id85=&id86=&id87=&id88=&id89=&id90=&id91=&id92=&id93=&id94=&id95=&id96=&id97=&id98=&id99=&id100=&id101=&id102=&id103=&id104=&id105=&id106=&id107=&id108=&id109=&id110=&id111=&id112=&id113=&id114=&id115=&id116=&id117=&id118=&id119=&id120=&id121=&id122=&id123=&id124=&id125=&id126=&id127=&id128=&id129=&id130=&id131=&id132=&id133=&id134=&id135=&id136=&id137=&id138=&id139=&id140=&id141=&id142=&id143=&id144=&id145=&id146=&id147=&id148=&id149=&id150=&id151=&id152=&id153=&id154=&id155=&id156=&id157=&id158=&id159=&id160=&id161=&id162=&id163=&id164=&id165=&id166=&id167=&id168=&id169=&id170=&id171=&id172=&id173=&id174=&id175=&id176=&id177=&id178=&id179=&id180=&id181=&id182=&id183=&id184=&id185=&id186=&id187=&id188=&id189=&id190=&id191=&id192=&id193=&id194=&id195=&id196=&id197=&id198=&id199=&id200=&id201=&id202=&id203=&id204=&id205=&id206=&id207=&id208=&id209=&id210=&id211=&id212=&id213=&id214=&id215=&id216=&id217=&id218=&id219=&id220=&id221=&id222=&id223=&id224=&id225=&id226=&id227=&id228=&id229=&id230=&id231=&id232=&id233=&id234=&id235=&id236=&id237=&id238=&id239=&id240=&id241=&id242=&id243=&id244=&id245=&id246=&id247=&id248=&id249=&id250=&id251=&id252=&id253=&id254=&id255=&id256=&id257=&id258=&id259=&id260=&id261=&id262=&id263=&id264=&id265=&id266=&id267=&id268=&id269=&id270=&id271=&id272=&id273=&id274=&id275=&id276=&id277=&id278=&id279=&id280=&id281=&id282=&id283=&id284=&id285=&id286=&id287=&id288=&id289=&id290=&id291=&id292=&id293=&id294=&id295=&id296=&id297=&id298=&id299=&id300=&id301=&id302=&id303=&id304=&id305=&id306=&id307=&id308=&id309=&id310=&id311=&id312=&id313=&id314=&id315=&id316=&id317=&id318=&id319=&id320=&id321=&id322=&id323=&id324=&id325=&id326=&id327=&id328=&id329=&id330=&id331=&id332=&id333=&id334=&id335=&id336=&id337=&id338=&id339=&id340=&id341=&id342=&id343=&id344=&id345=&id346=&id347=&id348=&id349=&id350=&id351=&id352=&id353=&id354=&id355=&id356=&id357=&id358=&id359=&id360=&id361=&id362=&id363=&id364=&id365=&id366=&id367=&id368=&id369=&id370=&id371=&id372=&id373=&id374=&id375=&id376=&id377=&id378=&id379=&id380=&id381=&id382=&id383=&id384=&id385=&id386=&id387=&id388=&id389=&id390=&id391=&id392=&id393=&id394=&id395=&id396=&id397=&id398=&id399=&id400=&id401=&id402=&id403=&id404=&id405=&id406=&id407=&id408=&id409=&id410=&id411=&id412=&id413=&id414=&id415=&id416=&id417=&id418=&id419=&id420=&id421=&id422=&id423=&id424=&id425=&id426=&id427=&id428=&id429=&id430=&id431=&id432=&id433=&id434=&id435=&id436=&id437=&id438=&id439=&id440=&id441=&id442=&id443=&id444=&id445=&id446=&id447=&id448=&id449=&id450=&id451=&id452=&id453=&id454=&id455=&id456=&id457=&id458=&id459=&id460=&id461=&id462=&id463=&id464=&id465=&id466=&id467=&id468=&id469=&id470=&id471=&id472=&id473=&id474=&id475=&id476=&id477=&id478=&id479=&id480=&id481=&id482=&id483=&id484=&id485=&id486=&id487=&id488=&id489=&id490=&id491=&id492=&id493=&id494=&id495=&id496=&id497=&id498=&id499=&id500=&id501=&id502=&id503=&id504=&id505=&id506=&id507=&id508=&id509=&id510=&id511=&id512=&id513=&id514=&id515=&id516=&id517=&id518=&id519=&id520=&id521=&id522=&id523=&id524=&id525=&id526=&id527=&id528=&id529=&id530=&id531=&id532=&id533=&id534=&id535=&id536=&id537=&id538=&id539=&id540=&id541=&id542=&id543=&id544=&id545=&id546=&id547=&id548=&id549=&id550=&id551=&id552=&id553=&id554=&id555=&id556=&id557=&id558=&id559=&id560=&id561=&id562=&id563=&id564=&id565=&id566=&id567=&id568=&id569=&id570=&id571=&id572=&id573=&id574=&id575=&id576=&id577=&id578=&id579=&id580=&id581=&id582=&id583=&id584=&id585=&id586=&id587=&id588=&id589=&id590=&id591=&id592=&id593=&id594=&id595=&id596=&id597=&id598=&id599=&id600=&id601=&id602=&id603=&id604=&id605=&id606=&id607=&id608=&id609=&id610=&id611=&id612=&id613=&id614=&id615=&id616=&id617=&id618=&id619=&id620=&id621=&id622=&id623=&id624=&id625=&id626=&id627=&id628=&id629=&id630=&id631=&id632=&id633=&id634=&id635=&id636=&id637=&id638=&id639=&id640=&id641=&id642=&id643=&id644=&id645=&id646=&id647=&id648=&id649=&id650=&id651=&id652=&id653=&id654=&id655=&id656=&id657=&id658=&id659=&id660=&id661=&id662=&id663=&id664=&id665=&id666=&id667=&id668=&id669=&id670=&id671=&id672=&id673=&id674=&id675=&id676=&id677=&id678=&id679=&id680=&id681=&id682=&id683=&id684=&id685=&id686=&id687=&id688=&id689=&id690=&id691=&id692=&id693=&id694=&id695=&id696=&id697=&id698=&id699=&id700=&id701=&id702=&id703=&id704=&id705=&id706=&id707=&id708=&id709=&id710=&id711=&id712=&id713=&id714=&id715=&id716=&id717=&id718=&id719=&id720=&id721=&id722=&id723=&id724=&id725=&id726=&id727=&id728=&id729=&id730=&id731=&id732=&id733=&id734=&id735=&id736=&id737=&id738=&id739=&id740=&id741=&id742=&id743=&id744=&id745=&id746=&id747=&id748=&id749=&id750=&id751=&id752=&id753=&id754=&id755=&id756=&id757=&id758=&id759=&id760=&id761=&id762=&id763=&id764=&id765=&id766=&id767=&id768=&id769=&id770=&id771=&id772=&id773=&id774=&id775=&id776=&id777=&id778=&id779=&id780=&id781=&id782=&id783=&id784=&id785=&id786=&id787=&id788=&id789=&id790=&id791=&id792=&id793=&id794=&id795=&id796=&id797=&id798=&id799=&id800=&id801=&id802=&id803=&id804=&id805=&id806=&id807=&id808=&id809=&id810=&id811=&id812=&id813=&id814=&id815=&id816=&id817=&id818=&id819=&id820=&id821=&id822=&id823=&id824=&id825=&id826=&id827=&id828=&id829=&id830=&id831=&id832=&id833=&id834=&id835=&id836=&id837=&id838=&id839=&id840=&id841=&id842=&id843=&id844=&id845=&id846=&id847=&id848=&id849=&id850=&id851=&id852=&id853=&id854=&id855=&id856=&id857=&id858=&id859=&id860=&id861=&id862=&id863=&id864=&id865=&id866=&id867=&id868=&id869=&id870=&id871=&id872=&id873=&id874=&id875=&id876=&id877=&id878=&id879=&id880=&id881=&id882=&id883=&id884=&id885=&id886=&id887=&id888=&id889=&id890=&id891=&id892=&id893=&id894=&id895=&id896=&id897=&id898=&id899=&id900=&id901=&id902=&id903=&id904=&id905=&id906=&id907=&id908=&id909=&id910=&id911=&id912=&id913=&id914=&id915=&id916=&id917=&id918=&id919=&id920=&id921=&id922=&id923=&id924=&id925=&id926=&id927=&id928=&id929=&id930=&id931=&id932=&id933=&id934=&id935=&id936=&id937=&id938=&id939=&id940=&id941=&id942=&id943=&id944=&id945=&id946=&id947=&id948=&id949=&id950=&id951=&id952=&id953=&id954=&id955=&id956=&id957=&id958=&id959=&id960=&id961=&id962=&id963=&id964=&id965=&id966=&id967=&id968=&id969=&id970=&id971=&id972=&id973=&id974=&id975=&id976=&id977=&id978=&id979=&id980=&id981=&id982=&id983=&id984=&id985=&id986=&id987=&id988=&id989=&id990=&id991=&id992=&id993=&id994=&id995=&id996=&id997=&id998=&id999	Public	Builds a road graph with the selected cost function (Section 11.4)
GET	/analytics/overview	AUTHORITY	KPI totals for the dashboard overview screen
GET	/analytics/health-score	AUTHORITY	Stretch: City Health Score per ward
GET	/analytics/heatmap	AUTHORITY	Stretch: point data for the Leaflet.heat layer

Example: POST /reports request/response

```
// Request (multipart/form-data)
{
  "photo": "<binary>",
  "description": "Large pothole near the main gate, cars swerving to avoid it",
  "lat": 23.1091,
  "lng": 72.5343
}
```

```
// Response
{
  "reportId": 4821,
  "clusterId": 117,
  "clusterStatus": "REPORTED",
  "aiClassification": {
    "issueType": "POTHOLE",
    "severity": 8,
    "department": "ROAD"
  },
  "mergedIntoExistingCluster": true,
  "clusterReportCount": 21,
  "priorityScore": 27.4
}
```

Example: GET /route response

```
{
  "mode": "safest",
  "distanceKm": 6.1,
  "etaMinutes": 14,
  "detourVsFastestPercent": 53,
  "flaggedStretchesAvoided": 1,
  "path": [
    { "lat": 23.1091, "lng": 72.5343, "nodeId": "n1" },
    { "lat": 23.1102, "lng": 72.5361, "nodeId": "n5" }
  ]
}
```

10. Step-by-Step Build Plan

Instructions for the AI executing this plan: Implement one step at a time. Do not jump ahead. After each step, the project should compile/run before moving to the next. Steps 1-16 are Must-Ship (Section 3.1); Steps 17-20+ are Stretch (Section 3.2) and can be attempted in any order or skipped entirely if time runs out.

STEP 1 — Scaffold the repository and Spring Boot backend

Goal: Create the repo skeleton from Section 6.1 and a running “hello world” Spring Boot app with PostgreSQL configured.

Files to create: - backend/pom.xml - backend/src/main/java/com/civicpulse/CivicPulseApplication
- backend/src/main/resources/application.yml - .env.example, .gitignore (root)

```
<!-- backend/pom.xml (key dependencies only) -->
<dependencies>
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-web</artifactId>
  </dependency>
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-data-jpa</artifactId>
  </dependency>
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-security</artifactId>
  </dependency>
  <dependency>
    <groupId>org.postgresql</groupId>
    <artifactId>postgresql</artifactId>
    <scope>runtime</scope>
  </dependency>
  <dependency>
    <groupId>io.jsonwebtoken</groupId>
    <artifactId>jjwt-api</artifactId>
    <version>0.11.5</version>
  </dependency>
</dependencies>
```

```
// backend/src/main/java/com/civicpulse/CivicPulseApplication.java
package com.civicpulse;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class CivicPulseApplication {
    public static void main(String[] args) {
        SpringApplication.run(CivicPulseApplication.class, args);
    }
}
```

```
# backend/src/main/resources/application.yml
spring:
```

```

datasource:
  url: ${DB_URL:jdbc:postgresql://localhost:5432/civimpulse}
  username: ${DB_USER:postgres}
  password: ${DB_PASSWORD:postgres}
jpa:
  hibernate:
    ddl-auto: update
    show-sql: true
server:
  port: 8080
civimpulse:
  jwt-secret: ${JWT_SECRET:change-me-in-env}
  gemini-api-key: ${GEMINI_API_KEY:}

```

Verify: `mvn spring-boot:run` starts the server on port 8080 with no errors.

STEP 2 — Data model

Goal: Create the JPA entities and repositories matching Section 8's schema.

Files: `model/User.java`, `model/Worker.java`, `model/IssueCluster.java`, `model/Report.java`, `model/Resolution.java`, and matching repository `*Repository.java` interfaces extending `JpaRepository`.

```

// backend/src/main/java/com/civimpulse/model/IssueCluster.java
package com.civimpulse.model;

import jakarta.persistence.*;
import java.time.LocalDateTime;

@Entity
@Table(name = "issue_clusters")
public class IssueCluster {
    @Id @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String issueType;
    private String department;
    private Integer severity;
    private Double centroidLat;
    private Double centroidLng;
    private String nearestEdgeId;
    private String status = "REPORTED";
    private Double priorityScore = 0.0;
    private Integer reportCount = 1;
    private Long assignedWorkerId;
    private LocalDateTime firstReportedAt = LocalDateTime.now();
    private LocalDateTime lastReportedAt = LocalDateTime.now();
    private LocalDateTime resolvedAt;

    // getters and setters omitted for brevity – generate with your IDE
    // or ask the AI: "generate getters/setters for IssueCluster"
}

```

```

// backend/src/main/java/com/civimpulse/repository/IssueClusterRepository.java
package com.civimpulse.repository;

```

```
import com.civicpulse.model.IssueCluster;
import org.springframework.data.jpa.repository.JpaRepository;
import java.util.List;

public interface IssueClusterRepository extends JpaRepository<IssueCluster, Long> {
    List<IssueCluster> findByIssueTypeAndStatusNot(String issueType, String status);
    List<IssueCluster> findAllByOrderByPriorityScoreDesc();
}
```

Repeat the same pattern (entity + repository) for User, Worker, Report, Resolution using the columns in Section 8. **Verify:** app starts and Hibernate auto-creates all five tables (check with \dt in psql).

STEP 3 — Report ingestion endpoint (no AI yet, category is manually picked)

Goal: POST /api/reports accepts a report, stores it, and creates a new cluster (dedupe comes in Step 8).

```
// backend/src/main/java/com/civicpulse/controller/ReportController.java
package com.civicpulse.controller;

import com.civicpulse.dto.ReportRequest;
import com.civicpulse.dto.ReportResponse;
import com.civicpulse.service.ReportService;
import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/api/reports")
public class ReportController {

    private final ReportService reportService;
    public ReportController(ReportService reportService) { this.reportService =
        ↪ reportService; }

    @PostMapping
    public ReportResponse submitReport(@RequestBody ReportRequest request) {
        return reportService.submitReport(request);
    }

    @GetMapping("/mine")
    public Object myReports(@RequestParam Long citizenId) {
        return reportService.findByCitizen(citizenId);
    }
}
```

```
// backend/src/main/java/com/civicpulse/dto/ReportRequest.java
package com.civicpulse.dto;

public class ReportRequest {
    public Long citizenId;
    public String photoUrl;           // for hackathon speed, upload photo to a simple
        ↪ storage                      // endpoint first (Step 3b) and pass the URL here
}
```

```

public String description;
public String issueType;      // manually picked category; overwritten if AI
    ↪ classification (Step 13) is on
public Double lat;
public Double lng;
}

```

The `ReportService.submitReport(...)` method for now: save a `Report` row, create a matching `IssueCluster` row with `reportCount = 1`, return the response. **This gets replaced in Step 8** to run dedupe first.

Step 3b (photo upload): for hackathon speed, add a simple `POST /api/uploads` that accepts multipart file data and stores it either to local disk (`/uploads/{uuid}.jpg`, served statically) or to a free-tier object store (Cloudinary/Supabase Storage both work without a paid key). Return the resulting URL for the citizen app to pass into `ReportRequest.photoUrl`.

Verify: `curl -X POST localhost:8080/api/reports -H "Content-Type: application/json" -d '{...}'` returns a `clusterId`.

STEP 4 — Basic dashboard read endpoints

Goal: `GET /api/clusters` (sorted by priority) and `GET /api/clusters/{id}` so the dashboard has something to render even before scoring/dedupe exist.

```

// backend/src/main/java/com/civicpulse/controller/ClusterController.java
package com.civicpulse.controller;

import com.civicpulse.model.IssueCluster;
import com.civicpulse.repository.IssueClusterRepository;
import org.springframework.web.bind.annotation.*;
import java.util.List;

@RestController
@RequestMapping("/api/clusters")
public class ClusterController {
    private final IssueClusterRepository repo;
    public ClusterController(IssueClusterRepository repo) { this.repo = repo; }

    @GetMapping
    public List<IssueCluster> all() { return repo.findAllByOrderByPriorityScoreDesc();
    ↪ }

    @GetMapping("/{id}")
    public IssueCluster one(@PathVariable Long id) { return
    ↪ repo.findById(id).orElseThrow(); }
}

```

Verify: the endpoint returns `[]` (empty list) once the DB is empty, and a populated list after a few Step-3 test submissions.

STEP 5 — Citizen App: report form + submit

Goal: A React screen matching Section 7.1 “Report an Issue” that posts to Step 3’s endpoint.

```
// frontend-citizen/src/pages/ReportForm.jsx
import { useState } from "react";

export default function ReportForm() {
  const [photo, setPhoto] = useState(null);
  const [description, setDescription] = useState("");
  const [issueType, setIssueType] = useState("POTHOLE");
  const [coords, setCoords] = useState(null);
  const [status, setStatus] = useState("");

  function captureLocation() {
    navigator.geolocation.getCurrentPosition(
      (pos) => setCoords({ lat: pos.coords.latitude, lng: pos.coords.longitude }),
      () => setStatus("Location permission denied – please enable GPS.")
    );
  }

  async function handleSubmit(e) {
    e.preventDefault();
    if (!coords) { setStatus("Please capture your location first."); return; }

    // Step 3b: upload the photo first, get back a URL
    const form = new FormData();
    form.append("file", photo);
    const uploadRes = await fetch("/api/uploads", { method: "POST", body: form });
    const { url: photoUrl } = await uploadRes.json();

    const res = await fetch("/api/reports", {
      method: "POST",
      headers: { "Content-Type": "application/json" },
      body: JSON.stringify({
        citizenId: 1, // replace with logged-in user id once auth (Step 17) is wired in
        photoUrl, description, issueType,
        lat: coords.lat, lng: coords.lng,
      }),
    });
    const data = await res.json();
    setStatus(
      data.mergedIntoExistingCluster
        ? `This issue already exists – ${data.clusterReportCount} citizens have  
↔ reported it now.`
        : "Report submitted! Thank you."
    );
  }

  return (
    <form onSubmit={handleSubmit} className="max-w-md mx-auto p-4 space-y-4">
      <input type="file" accept="image/*" capture="environment"
        onChange={(e) => setPhoto(e.target.files[0])} required />
      <select value={issueType} onChange={(e) => setIssueType(e.target.value)}
        className="w-full border p-2 rounded">
        <option value="POTHOLE">Pothole</option>
        <option value="STREETLIGHT">Streetlight</option>
        <option value="WATER">Water Leakage</option>
        <option value="GARBAGE">Garbage</option>
        <option value="SAFETY">Safety</option>
        <option value="OTHER">Other</option>
      </select>
    </form>
  );
}
```

```

<textarea value={description} onChange={(e) => setDescription(e.target.value)}
  placeholder="Optional description" className="w-full border p-2
  ↪ rounded" />
<button type="button" onClick={captureLocation} className="border px-3 py-2
  ↪ rounded">
  {coords ? `Location captured ✓ (${coords.lat.toFixed(4)},
  ↪ ${coords.lng.toFixed(4)})` : "Capture Location"}
</button>
<button type="submit" className="bg-green-600 text-white px-4 py-2 rounded
  ↪ w-full">
  Submit Report
</button>
{status} && <p className="text-sm text-gray-700">{status}</p>
</form>
);
}

```

Verify: submitting from the browser creates a row visible via Step 4's GET /api/clusters.

STEP 6 — Citizen App: map view with color-coded pins

Goal: Section 7.1 “Map / Track” screen.

```

// frontend-citizen/src/pages/IssueMap.jsx
import { useEffect, useState } from "react";
import { MapContainer, TileLayer, CircleMarker, Popup } from "react-leaflet";

const PRIORITY_COLOR = (score) => (score >= 20 ? "red" : score >= 8 ? "orange" :
  ↪ "green");

export default function IssueMap() {
  const [clusters, setClusters] = useState([]);

  useEffect(() => {
    fetch("/api/clusters").then((r) => r.json()).then(setClusters);
  }, []);

  return (
    <MapContainer center={[23.1091, 72.5343]} zoom={15} style={{ height: "80vh" }}>
      <TileLayer url="https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png" />
      {clusters.map((c) => (
        <CircleMarker key={c.id} center={[c.centroidLat, c.centroidLng]}
          radius={8} pathOptions={{ color: PRIORITY_COLOR(c.priorityScore)
          ↪ }}>
          <Popup>
            <b>{c.issueType}</b><br />
            Priority: {c.priorityScore.toFixed(1)}<br />
            Reports: {c.reportCount}<br />
            Status: {c.status}
          </Popup>
        </CircleMarker>
      ))}
    </MapContainer>
  );
}

```

Verify: pins render at the correct GPS location and color changes as priorityScore changes (test by manually editing a row's priority_score in the DB and refreshing).

STEP 7 — Authority Dashboard: overview + issue queue

Goal: Section 7.2 screens, reading from Step 4's endpoints.

```
// frontend-dashboard/src/pages/Overview.jsx
import { useEffect, useState } from "react";

export default function Overview() {
  const [clusters, setClusters] = useState([]);
  useEffect(() => { fetch("/api/clusters").then((r) => r.json()).then(setClusters); },
    []);

  const counts = clusters.reduce((acc, c) => {
    acc.total++; acc[c.status] = (acc[c.status] || 0) + 1; return acc;
  }, { total: 0 });

  return (
    <div className="p-6">
      <div className="grid grid-cols-4 gap-4 mb-6">
        <KpiCard label="Total Issues" value={counts.total} />
        <KpiCard label="Pending" value={{(counts.REPORTED || 0) + (counts.VERIFIED || 0)}} />
        <KpiCard label="In Progress" value={counts.IN_PROGRESS || 0} />
        <KpiCard label="Resolved" value={counts.RESOLVED || 0} />
      </div>
      <table className="w-full border-collapse">
        <thead><tr className="text-left border-b">
          <th>Type</th><th>Priority</th><th>Reports</th><th>Status</th><th>Action</th>
        </tr></thead>
        <tbody>
          {clusters.map((c) => (
            <tr key={c.id} className="border-b">
              <td>{c.issueType}</td>
              <td>{c.priorityScore.toFixed(1)}</td>
              <td>{c.reportCount}</td>
              <td>{c.status}</td>
              <td><button className="text-blue-600 underline">Assign
                Worker</button></td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}

function KpiCard({ label, value }) {
  return <div className="border rounded p-4 text-center">
    <div className="text-2xl font-bold">{value}</div>
    <div className="text-sm text-gray-500">{label}</div>
  </div>;
}
```

Verify: KPI counts match the number of rows in the DB grouped by status.

STEP 8 — Duplicate detection / clustering engine

Goal: Replace Step 3’s “always create a new cluster” logic with real dedupe: a new report merges into an existing open cluster if it’s the same `issueType`, within `DEDUPE_RADIUS_METERS` (default 50m), and within `DEDUPE_TIME_WINDOW_DAYS` (default 3 days) of the cluster’s `lastReportedAt`. Full algorithm and code are in **Section 11.1** — implement that code now as `DuplicateDetectionService`, then wire it into `ReportService.submitReport(...)`:

```
// backend/src/main/java/com/civicpulse/service/ReportService.java (updated
↪ submitReport)
public ReportResponse submitReport(ReportRequest req) {
    Report report = new Report();
    report.setCitizenId(req.citizenId);
    report.setPhotoUrl(req.photoUrl);
    report.setDescription(req.description);
    report.setLat(req.lat);
    report.setLng(req.lng);

    IssueCluster cluster = duplicateDetectionService.findMatchingCluster(req.issueType,
↪ req.lat, req.lng);
    boolean merged = cluster != null;
    if (!merged) {
        cluster = new IssueCluster();
        cluster.setIssueType(req.issueType);
        cluster.setDepartment(departmentFor(req.issueType));
        cluster.setSeverity(defaultSeverityFor(req.issueType));
        cluster.setCentroidLat(req.lat);
        cluster.setCentroidLng(req.lng);
    } else {
        cluster.setReportCount(cluster.getReportCount() + 1);
        cluster.setLastReportedAt(LocalDate.now());
        // recompute centroid as running average, keeps the pin near the true cluster
        ↪ center
        cluster.setCentroidLat((cluster.getCentroidLat() * (cluster.getReportCount() -
↪ 1) + req.lat) / cluster.getReportCount());
        cluster.setCentroidLng((cluster.getCentroidLng() * (cluster.getReportCount() -
↪ 1) + req.lng) / cluster.getReportCount());
    }
    cluster = clusterRepository.save(cluster);
    report.setClusterId(cluster.getId());
    report = reportRepository.save(report);

    // Step 9 wires in real scoring here:
    double priority = priorityScoringService.computePriority(cluster);
    cluster.setPriorityScore(priority);
    clusterRepository.save(cluster);

    return new ReportResponse(report.getId(), cluster.getId(), cluster.getStatus(),
        merged, cluster.getReportCount(), priority);
}
```

Verify: submit two reports of the same category within 50m of each other (use nearby test coordinates) — the second call should return `mergedIntoExistingCluster: true` and `clusterReportCount: 2`, and only one row should exist in `issue_clusters`.

STEP 9 — Priority scoring engine

Goal: Implement the formula from **Section 11.2** as `PriorityScoringService.computePriority(IssueCluster)`, already wired into Step 8 above. For Step 9, the `criticality_multiplier` can temporarily return a hardcoded 1.0 for every cluster — it becomes real once Step 10/11 (the road graph + centrality) exist.

Verify: manually create clusters with different severities/report counts/ages via the API and confirm `priorityScore` in the response matches a hand-calculated value from the formula.

STEP 10 — Offline road graph construction (Python, run once)

Goal: Produce a small JSON file describing the road network for your demo area (campus or one ward), with a precomputed centrality score per edge, that the Java backend loads at startup. Full script is in **Section 11.3**. Run it once before the hackathon demo (and again if you change the bounding box):

```
cd graph-preprocessing
pip install -r requirements.txt
python build_graph.py --place "Nirma University, Ahmedabad, India" --out
↪ ../backend/src/main/resources/road_graph.json
```

Verify: `road_graph.json` exists and contains a non-empty "nodes" and "edges" array, and each edge has a `centralityScore` between 0 and 1.

STEP 11 — Load the graph into the backend + Dijkstra/A* routing

Goal: At Spring Boot startup, load `road_graph.json` into an in-memory graph (`RoadGraphService`), and expose GET `/api/route`. Full graph model and shortest-path code are in **Section 11.4**.

```
// backend/src/main/java/com/civicpulse/controller/RouteController.java
package com.civicpulse.controller;

import com.civicpulse.graph.RoadGraphService;
import com.civicpulse.graph.RouteResult;
import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/api/route")
public class RouteController {
    private final RoadGraphService graphService;

    public RouteController(RoadGraphService graphService) { this.graphService =
        graphService; }

    @GetMapping
    public RouteResult route(@RequestParam double fromLat, @RequestParam double fromLng,
                             @RequestParam double toLat, @RequestParam double toLng,
                             @RequestParam(defaultValue = "fastest") String mode) {
        return graphService.findRoute(fromLat, fromLng, toLat, toLng, mode);
    }
}
```

Verify: GET `/api/route?fromLat=...&fromLng=...&toLat=...&toLng=...&mode=fastest` returns a path; changing `mode=safest` returns a *different* path once at least one open, high-

severity cluster sits on an edge of the fastest path (test by submitting a report near one of the fastest route's edges, then re-querying).

STEP 12 — Citizen App: fastest/safest route toggle (first Stretch item, but do it right after Step 11 while the graph is fresh in your mind)

```
// frontend-citizen/src/pages/RoutePlanner.jsx
import { useState } from "react";
import { MapContainer, TileLayer, Polyline, Marker } from "react-leaflet";

export default function RoutePlanner() {
  const [from, setFrom] = useState(null);
  const [to, setTo] = useState(null);
  const [route, setRoute] = useState(null);
  const [mode, setMode] = useState("fastest");

  async function fetchRoute(selectedMode) {
    setMode(selectedMode);
    const params = new URLSearchParams({
      fromLat: from.lat, fromLng: from.lng, toLat: to.lat, toLng: to.lng, mode:
        ↪ selectedMode,
    });
    const res = await fetch(`/api/route?${params}`);
    setRoute(await res.json());
  }

  return (
    <div>
      <div className="flex gap-2 p-2">
        <button onClick={() => fetchRoute("fastest")}
          className={mode === "fastest" ? "font-bold underline" : ""}>Fastest
          ↪ Route</button>
        <button onClick={() => fetchRoute("safest")}
          className={mode === "safest" ? "font-bold underline" : ""}>Safest
          ↪ Route</button>
      </div>
      {route && (
        <p className="px-2 text-sm">
          {mode === "fastest"
            ? `Fastest: ${route.distanceKm} km, passes ${route.flaggedStretchesAvoided}
              ↪ === 0 ? "0" : "≥1"} flagged stretch`
            : `Safest: ${route.distanceKm} km (+${route.detourVsFastestPercent}%),
              ↪ avoids all flagged stretches`}
          </p>
        )}
      <MapContainer center={[23.1091, 72.5343]} zoom={15} style={{ height: "70vh" }}>
        <TileLayer url="https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png" />
        {route && (
          <Polyline positions={route.path.map((p) => [p.lat, p.lng])}
            pathOptions={{ color: mode === "safest" ? "green" : "orange" }} />
        )}
      </MapContainer>
    </div>
  );
}
```

Verify: toggling between Fastest/Safest visibly redraws a different colored line on the map, matching the toy-network demo described in Section 11.4.

STEP 13 — Gemini classification (Stretch, but do it before worker assignment if time allows)

Goal: Replace the manual category picker's authority with an AI call. Full prompt design is in Section 11.5.

```
// backend/src/main/java/com/civicpulse/service/GeminiClassificationService.java
package com.civicpulse.service;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Service;
import org.springframework.web.client.RestTemplate;
import java.util.Map;

@Service
public class GeminiClassificationService {

    @Value("${civicpulse.gemini-api-key}")
    private String apiKey;

    private final RestTemplate restTemplate = new RestTemplate();

    public Classification classify(String photoUrl, String description) {
        if (apiKey == null || apiKey.isBlank()) {
            return keywordFallback(description); // demo-safety net, see Section 11.5
        }
        try {
            String url =
                ↪ "https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-
                ↪ flash:generateContent?key=" +
                ↪ apiKey;
            String prompt = ""
                ↪ "Classify this civic issue report. Respond ONLY with JSON in this exact
                ↪ shape:
                ↪ {
                ↪   "issueType": "POTHOLE|STREETLIGHT|WATER|GARBAGE|SAFETY|OTHER",
                ↪   "severity": <integer 1-10>, "department":
                ↪ "ROAD|ELECTRICAL|WATER|SANITATION|SAFETY"}
                ↪ Description: %s
                ↪ Photo URL: %s
                ↪ ""formatted(description == null ? "(none provided)" : description,
                ↪ photoUrl);

            Map<String, Object> body = Map.of("contents",
                ↪ new Object[]{ Map.of("parts", new Object[]{ Map.of("text", prompt)
                ↪ }) }]);
            Map response = restTemplate.postForObject(url, body, Map.class);
            String text = extractText(response); // parse Gemini's nested response
            ↪ structure
            return parseJson(text);
        } catch (Exception e) {
            return keywordFallback(description); // never let a flaky API break the demo
        }
    }
}
```

```

private Classification keywordFallback(String description) {
    String d = description == null ? "" : description.toLowerCase();
    if (d.contains("pothole") || d.contains("road")) return new
        ↪ Classification("POTHOLE", 7, "ROAD");
    if (d.contains("light")) return new Classification("STREETLIGHT", 5,
        ↪ "ELECTRICAL");
    if (d.contains("water") || d.contains("leak")) return new
        ↪ Classification("WATER", 6, "WATER");
    if (d.contains("garbage") || d.contains("trash")) return new
        ↪ Classification("GARBAGE", 4, "SANITATION");
    return new Classification("OTHER", 3, "SAFETY");
}

// extractText() / parseJson() helper methods: pull
↪ response.candidates[0].content.parts[0].text
// and org.json / Jackson-parse the JSON string it contains.

public record Classification(String issueType, int severity, String department) {}
}

```

Verify: with a real GEMINI_API_KEY in .env, submitting a report returns a sensible classification; with the key removed, the same endpoint still works via the keyword fallback (test both paths — judges may ask “what happens when the API is down?”).

STEP 14 — Authority Dashboard: verify + assign controls

Goal: Wire up PATCH /clusters/{id}/verify, PATCH /clusters/{id}/priority, PATCH /clusters/{id}/assign on the backend and corresponding buttons/modals in Overview.jsx (Step 7) so an authority user can correct the AI, override priority, and assign a worker.

```

// backend/src/main/java/com/civicpulse/controller/ClusterController.java (additions)
@PatchMapping("/{id}/verify")
public IssueCluster verify(@PathVariable Long id, @RequestBody VerifyRequest body) {
    return clusterService.verifyClassification(id, body.issueType, body.severity,
        ↪ body.department);
}

@PatchMapping("/{id}/priority")
public IssueCluster overridePriority(@PathVariable Long id, @RequestBody
    ↪ PriorityOverrideRequest body) {
    return clusterService.overridePriority(id, body.priorityScore);
}

@PatchMapping("/{id}/assign")
public IssueCluster assign(@PathVariable Long id, @RequestBody AssignRequest body) {
    return clusterService.assignWorker(id, body.workerId);
}

```

Verify: clicking “Assign Worker” in the dashboard updates assigned_worker_id in the DB and the cluster’s status moves to ASSIGNED.

STEP 15 — Worker assignment engine (Stretch)

Goal: GET /clusters/{id}/suggest-worker ranks eligible workers using the scoring function in **Section 11.6**, then the dashboard’s “Assign Worker” button calls this endpoint first and pre-fills the top suggestion (authority can still override manually — never auto-assign without a human able to see/override it, this matters for the “authority can verify/reassign” pitch point).

```
// backend/src/main/java/com/civicpulse/controller/ClusterController.java (addition)
@GetMapping("/{id}/suggest-worker")
public List<WorkerScore> suggestWorker(@PathVariable Long id) {
    return workerAssignmentService.rankCandidates(id);
}
```

Verify: the ranked list places a nearer, less-loaded, correct-domain worker above a farther, overloaded, or wrong-domain one — test with at least 3 seeded workers of different departments/workloads.

STEP 16 — Worker App: task list + navigate + resolve

Goal: Section 7.3 screens.

```
// frontend-worker/src/pages/TaskList.jsx
import { useEffect, useState } from "react";

export default function TaskList({ workerId }) {
    const [tasks, setTasks] = useState([]);
    useEffect(() => {
        fetch(`/api/workers/${workerId}/tasks`).then((r) => r.json()).then(setTasks);
    }, [workerId]);

    function navigate(task) {
        // deep-link into the device's default maps app — no API key needed
        window.open(`https://www.google.com/maps/dir/?api=1&destination=${task.lat},${task.lng}`,
            task.lng,
            "_blank");
    }

    return (
        <div className="p-4 space-y-3">
            {tasks.map((t) => (
                <div key={t.id} className="border rounded p-3">
                    <div className="flex justify-between">
                        <span className={t.priorityScore >= 20 ? "text-red-600 font-bold" : "text-orange-600"}>
                            {t.priorityScore >= 20 ? "[HIGH] HIGH" : "[MED] MEDIUM"} — {t.issueType}
                        </span>
                        <span>{t.distanceKm.toFixed(1)} km away</span>
                    </div>
                    <button onClick={() => navigate(t)} className="mt-2 bg-blue-600 text-white px-3 py-1 rounded">
                        Navigate
                    </button>
                </div>
            ))}
        </div>
    );
}
```

```
}
```

```
// backend/src/main/java/com/civicpulse/controller/ClusterController.java (resolve  
↪ endpoint)  
@PatchMapping("/{id}/resolve")  
public IssueCluster resolve(@PathVariable Long id, @RequestBody ResolveRequest body) {  
    return clusterService.resolve(id, body.workerId, body.afterPhotoUrl, body.notes);  
}
```

clusterService.resolve(...) should: create a Resolution row, set cluster.status = "RESOLVED", resolvedAt = now(), decrement worker.activeTaskCount, and (in a full build) push a notification to the citizen who first reported the issue.

Verify: marking a task resolved moves it out of the worker's active task list and the citizen's tracking screen (Step 6) shows status: RESOLVED on next refresh.

At this point, Steps 1-16 form a complete, demoable Must-Ship MVP satisfying the mandatory UI requirement and covering all three interfaces. Everything from Step 17 onward is optional polish, attempted top-down per the payoff ranking in Section 3.2.

STEP 17 — Authentication & role-based access

Add Spring Security with JWT (three roles: CITIZEN, AUTHORITY, WORKER), replace the hard-coded citizenId: 1 in Step 5's frontend with the logged-in user's real ID from the JWT, and lock down /clusters/** mutation endpoints to AUTHORITY only and /workers/** to WORKER only via @PreAuthorize.

STEP 18 — Heatmap layer (Stretch #6, Section 3.2)

Add leaflet.heat to the dashboard map; feed it GET /analytics/heatmap (a flat list of {lat, lng, weight} where weight = priorityScore) — roughly 20-30 minutes of work once cluster data exists.

STEP 19 — City Health Score (Stretch #3, Section 3.2)

Implement per Section 12.2; add a GET /analytics/health-score endpoint and a simple ranked list or choropleth on the dashboard's Analytics screen.

STEP 20+ — Remaining stretch features

Implement any of Section 12's remaining specs (predictive alerts, before/after verification, reporter trust score, SLA tracking, reward system, civic copilot, voice intake, route safety score) in the priority order given in Section 3.2, stopping whenever you run out of hours — none of them are load-bearing for the Must-Ship demo.

11. Algorithm Deep Dives

11.1 Duplicate / Cluster Detection

Idea: No ML needed. Two reports of the same category within a distance threshold and a time window are the same real-world issue. Distance between two GPS points uses the **Haversine formula**:

$$a = \sin^2(\Delta\text{lat}/2) + \cos(\text{lat1}) \cdot \cos(\text{lat2}) \cdot \sin^2(\Delta\text{lng}/2)$$
$$c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a})$$
$$\text{distance_meters} = R \cdot c \quad (R = \text{Earth's radius} \approx 6,371,000 \text{ m})$$

```
// backend/src/main/java/com/civimpulse/service/DuplicateDetectionService.java
package com.civimpulse.service;

import com.civimpulse.model.IssueCluster;
import com.civimpulse.repository.IssueClusterRepository;
import org.springframework.stereotype.Service;
import java.time.LocalDateTime;
import java.time.temporal.ChronoUnit;
import java.util.List;

@Service
public class DuplicateDetectionService {

    private static final double DEDUPE_RADIUS_METERS = 50.0;
    private static final long DEDUPE_TIME_WINDOW_DAYS = 3;
    private static final double EARTH_RADIUS_METERS = 6_371_000;

    private final IssueClusterRepository clusterRepository;
    public DuplicateDetectionService(IssueClusterRepository clusterRepository) {
        this.clusterRepository = clusterRepository;
    }

    public IssueCluster findMatchingCluster(String issueType, double lat, double lng) {
        List<IssueCluster> candidates =
            ↪ clusterRepository.findByIssueTypeAndStatusNot(issueType, "RESOLVED");
        for (IssueCluster c : candidates) {
            boolean withinTime = ChronoUnit.DAYS.between(c.getLastReportedAt(),
                ↪ LocalDateTime.now()) <= DEDUPE_TIME_WINDOW_DAYS;
            boolean withinRadius = haversineMeters(lat, lng, c.getCentroidLat(),
                ↪ c.getCentroidLng()) <= DEDUPE_RADIUS_METERS;
            if (withinTime && withinRadius) return c;
        }
        return null; // no match -> caller creates a new cluster
    }

    public static double haversineMeters(double lat1, double lng1, double lat2, double
        ↪ lng2) {
        double dLat = Math.toRadians(lat2 - lat1);
        double dLng = Math.toRadians(lng2 - lng1);
        double a = Math.sin(dLat / 2) * Math.sin(dLat / 2)
            + Math.cos(Math.toRadians(lat1)) * Math.cos(Math.toRadians(lat2))
            * Math.sin(dLng / 2) * Math.sin(dLng / 2);
        double c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
        return EARTH_RADIUS_METERS * c;
    }
}
```

Complexity: $O(k)$ per new report, where k = number of currently-open clusters of that issue type in the area — trivially fast for a hackathon-scale demo (dozens to low hundreds of clusters). At real city scale you’d add a spatial index (PostGIS ST_DWithin + a GIST index, or a geohash bucket) so you only compare against nearby candidates instead of every open cluster — worth mentioning in Q&A as the “how would this scale” answer.

Image-similarity extension (optional, higher risk): if two reports are within radius/time but you want extra confidence they’re the *same* pothole and not two different potholes 40m apart, you can additionally compare image embeddings (e.g., a simple perceptual hash, or a Gemini vision “are these the same defect?” call) — mention this as a possible Q&A answer even if you don’t implement it; the geometry+time heuristic alone already outperforms a plain report-count system.

11.2 Priority Scoring Formula

$\text{priority} = \text{severity_weight} \times \text{cluster_size} \times \text{recency_decay} \times \text{criticality_multiplier}$

Factor	Meaning	Example values
severity_weight	1-10, from AI classification or authority override	Pothole = 8, graffiti = 1
cluster_size	Number of individual reports merged into this civic issue	5 reports → 5
recency_decay	Recent clusters matter more; decays toward 0.3 over ~14 days	$\max(0.3, 1.0 - \text{daysSinceFirstReport} / 14.0)$
criticality_multiplier	Structural importance of the road segment, from graph centrality (Section 11.3)	Near a school/hospital/main road → 1.5-2.0; quiet lane → 1.0

```
// backend/src/main/java/com/civicpulse/service/PriorityScoringService.java
package com.civicpulse.service;

import com.civicpulse.graph.RoadGraphService;
import com.civicpulse.model.IssueCluster;
import org.springframework.stereotype.Service;
import java.time.temporal.ChronoUnit;

@Service
public class PriorityScoringService {

    private final RoadGraphService roadGraphService;
    public PriorityScoringService(RoadGraphService roadGraphService) {
        ↪ this.roadGraphService = roadGraphService; }

    public double computePriority(IssueCluster cluster) {
        double severityWeight = cluster.getSeverity();
        double clusterSize = cluster.getReportCount();
        long daysOpen =
            ↪ ChronoUnit.DAYS.between(cluster.getFirstReportedAt().toLocalDate(),
            ↪ java.time.LocalDate.now());
        double recencyDecay = Math.max(0.3, 1.0 - daysOpen / 14.0);
        double criticalityMultiplier = roadGraphService.criticalityMultiplierFor(
            cluster.getCentroidLat(), cluster.getCentroidLng());
```

```

        return severityWeight * clusterSize * recencyDecay * criticalityMultiplier;
    }
}

```

Worked example (matches the pitch in Section 1): a pothole (severity 3 on a 1-10 scale used loosely as “3 = moderate”) reported 5 times in the last 2 days (recencyDecay ≈ 1.0) near a school (criticalityMultiplier = 1.8) scores $3 \times 5 \times 1.0 \times 1.8 = 27$. A single graffiti report on a quiet lane (severity = 1, clusterSize = 1, criticalityMultiplier = 1.0) scores $1 \times 1 \times 1.0 \times 1.0 = 1$. That gap is your defensible answer to “why does the dashboard show this one first” — most competing report-count-only systems can’t say more than “it got the most votes.”

11.3 Road Graph Construction & Betweenness Centrality (offline, Python)

Why offline: Betweenness centrality (Brandes’ algorithm — repeated BFS/Dijkstra from every node) is $O(V \cdot E)$ or worse; fine to run **once** over a small campus/ward graph, but far too slow to run on every request. Precompute once, cache the result in the exported JSON, and the live backend just does $O(1)$ lookups.

```

# graph-preprocessing/build_graph.py
import argparse, json
import osmnx as ox
import networkx as nx

def main():
    parser = argparse.ArgumentParser()
    parser.add_argument("--place", required=True, help='e.g. "Nirma University,
    ↪ Ahmedabad, India"')
    parser.add_argument("--out", required=True)
    args = parser.parse_args()

    # Pull the real road network for the given place via Overpass (no API key needed)
    G = ox.graph_from_place(args.place, network_type="drive")

    # Undirected view for centrality; keep directed graph for routing distances if
    ↪ needed
    Gu = G.to_undirected()

    # Brandes' betweenness centrality -- NetworkX ships this, no need to hand-roll it
    edge centrality = nx.edge_betweenness centrality(Gu, weight="length",
    ↪ normalized=True)

    nodes_out = [{"id": str(n), "lat": data["y"], "lng": data["x"]} for n, data in
    ↪ G.nodes(data=True)]

    edges_out = []
    for u, v, data in G.edges(data=True):
        length_m = data.get("length", 50.0)
        base_travel_time_s = length_m / (data.get("maxspeed_ms", 8.0)) # ~29 km/h
    ↪ fallback
        centrality = edge centrality.get((u, v), edge centrality.get((v, u), 0.0))
        edges_out.append({
            "id": f"{u}-{v}",
            "from": str(u), "to": str(v),
            "lengthMeters": length_m,
            "baseTravelTimeSeconds": base_travel_time_s,

```

```

        "centralityScore": centrality,    # 0..1, precomputed once
    })

    with open(args.out, "w") as f:
        json.dump({"nodes": nodes_out, "edges": edges_out}, f)

    print(f"Wrote {len(nodes_out)} nodes and {len(edges_out)} edges to {args.out}")

if __name__ == "__main__":
    main()

```

```

# graph-preprocessing/requirements.txt
osmnx>=1.9
networkx>=3.2

```

Turning centralityScore into the priority formula’s criticality_multiplier: at load time, scale the 0–1 centrality score into a 1.0–2.0 multiplier: `criticalityMultiplier = 1.0 + centralityScore` (or apply a manual boost for edges tagged near a school/hospital in OSM’s amenity tags, if you want the exact “near school +2” flavor described in the pitch — see the note at the end of Section 11.4).

11.4 Risk-Weighted Shortest Path (Dijkstra / A*)

The trick: fastest and safest routes are the *same* shortest-path call with two different edge-cost functions.

```

cost_fastest(edge) = baseTravelTimeSeconds
cost_safest(edge)  = baseTravelTimeSeconds × (1 + openFlaggedReports(edge) × severityFactor)

```

```

// backend/src/main/java/com/civicpulse/graph/RoadGraphService.java
package com.civicpulse.graph;

import org.springframework.stereotype.Service;
import java.util.*;

@Service
public class RoadGraphService {

    private Map<String, Node> nodes = new HashMap<>();
    private Map<String, List<Edge>> adjacency = new HashMap<>();
    // loaded from road_graph.json at startup via @PostConstruct – see loadGraph() below

    public RouteResult findRoute(double fromLat, double fromLng, double toLat, double
    ↪ toLng, String mode) {
        String startId = nearestNode(fromLat, fromLng);
        String endId = nearestNode(toLat, toLng);

        // The ONLY difference between "fastest" and "safest": which cost function we
        ↪ hand to Dijkstra.
        java.util.function.ToDoubleFunction<Edge> costFn = mode.equals("safest")
            ? this::safestCost
            : this::fastestCost;

        return dijkstra(startId, endId, costFn, mode);
    }

    private double fastestCost(Edge e) {
        return e.baseTravelTimeSeconds;
    }

```

```

    }

    private double safestCost(Edge e) {
        double severityFactor = 0.3; // tunable constant
        return e.baseTravelTimeSeconds * (1 + e.openFlaggedReports * severityFactor);
    }

    private RouteResult dijkstra(String startId, String endId,
        java.util.function.ToDoubleFunction<Edge> costFn,
        String mode) {
        Map<String, Double> dist = new HashMap<>();
        Map<String, String> prev = new HashMap<>();
        PriorityQueue<String> pq = new PriorityQueue<>(Comparator.comparingDouble(n ->
            dist.getOrDefault(n, Double.MAX_VALUE)));
        dist.put(startId, 0.0);
        pq.add(startId);

        while (!pq.isEmpty()) {
            String u = pq.poll();
            if (u.equals(endId)) break;
            for (Edge e : adjacency.getOrDefault(u, List.of())) {
                double newDist = dist.get(u) + costFn.applyAsDouble(e);
                if (newDist < dist.getOrDefault(e.to, Double.MAX_VALUE)) {
                    dist.put(e.to, newDist);
                    prev.put(e.to, u);
                    pq.add(e.to);
                }
            }
        }
        return buildResult(startId, endId, dist, prev, mode);
    }

    // buildResult() walks `prev` backward from endId to startId to build the path,
    // and computes distanceKm / etaMinutes / flaggedStretchesAvoided /
    // by comparing against a fastest-mode run when mode == "safest".
    // ↪ detourVsFastestPercent

    public double criticalityMultiplierFor(double lat, double lng) {
        String nodeId = nearestNode(lat, lng);
        double maxCentrality = adjacency.getOrDefault(nodeId, List.of()).stream()
            .mapToDouble(e -> e.centraliityScore).max().orElse(0.0);
        return 1.0 + maxCentrality; // 1.0 (unimportant) .. 2.0 (highly central)
    }

    // nearestNode(), loadGraph() (@PostConstruct reading road_graph.json into
    // ↪ `nodes`/`adjacency`,
    // and updating each edge's openFlaggedReports whenever a cluster's nearestEdgeId
    // ↪ is set/resolved)
    // are straightforward I/O helpers – implement with standard Jackson JSON parsing.

    static class Node { String id; double lat, lng; }
    static class Edge {
        String to; double baseTravelTimeSeconds, centralityScore; int
        ↪ openFlaggedReports;
    }
}

```

Live-demo tip: build one small, deliberately obvious toy detour into your demo area (e.g., a

“shortcut lane” flagged unsafe) so the fastest-vs-safest toggle visibly picks a longer, greener path around it — exactly like the toy network described in the original pitch (fastest 4.0 km through a flagged stretch vs. safest 6.1 km, +53%, avoiding it entirely). A visible, understandable trade-off sells the algorithm far better than a subtle 5% difference on a real, busy street grid.

11.5 Gemini Classification Prompt Design

Rule of thumb from the pitch material, worth repeating to the AI implementing this: *don't let Gemini decide assignment — only classification/extraction.* Keep the prompt narrow and force strict JSON output so your own backend code (Sections 11.1–11.2, 11.6) makes every downstream decision:

Classify this civic issue report. Respond ONLY with JSON in this exact shape:
{"issueType": "POTHOLE|STREETLIGHT|WATER|GARBAGE|SAFETY|OTHER",
"severity": <integer 1-10>, "department": "ROAD|ELECTRICAL|WATER|SANITATION|SAFETY"}
Description: {description}
Photo URL: {photoUrl}

Always keep a keyword-based fallback (Step 13's keywordFallback(...)) so a flaky network or exhausted free-tier quota during the live demo never breaks the report flow — and be ready to explain this fallback in Q&A, since Rule 11 explicitly expects you to be able to discuss AI-tool usage and its limitations.

11.6 Worker Assignment Scoring

$\text{workerScore} = (\text{domainMatch} ? 1 : 0) \times \text{priorityWeight} \times (1 / (1 + \text{distanceKm})) \times (1 / (1 + \text{activeT}))$

- domainMatch: hard filter — a WATER-department worker never gets suggested for a POTHOLE (ROAD-department) issue.
- priorityWeight: the cluster's own priorityScore (Section 11.2) — higher-priority issues favor pulling in a worker even at slightly greater distance.
- Distance and workload are both **inverted** so nearer, less-loaded workers score higher.

```
// backend/src/main/java/com/civicpulse/service/WorkerAssignmentService.java
package com.civicpulse.service;
```

```
import com.civicpulse.model.IssueCluster;
import com.civicpulse.model.Worker;
import com.civicpulse.repository.WorkerRepository;
import com.civicpulse.repository.IssueClusterRepository;
import org.springframework.stereotype.Service;
import java.util.Comparator;
import java.util.List;
import java.util.stream.Collectors;
```

```
@Service
```

```
public class WorkerAssignmentService {
```

```
    private final WorkerRepository workerRepository;
```

```
    private final IssueClusterRepository clusterRepository;
```

```
    public WorkerAssignmentService(WorkerRepository workerRepository,
```

```
        IssueClusterRepository clusterRepository) {
```

```
        this.workerRepository = workerRepository;
```

```
        this.clusterRepository = clusterRepository;
```

```
    }
```

```

public List<WorkerScore> rankCandidates(Long clusterId) {
    IssueCluster cluster = clusterRepository.findById(clusterId).orElseThrow();
    List<Worker> eligible =
        ↪ workerRepository.findByDepartment(cluster.getDepartment());

    return eligible.stream()
        .map(w -> {
            double distanceKm = DuplicateDetectionService.haversineMeters(
                cluster.getCentroidLat(), cluster.getCentroidLng(),
                w.getCurrentLat(), w.getCurrentLng()) / 1000.0;
            double score = cluster.getPriorityScore()
                * (1.0 / (1.0 + distanceKm))
                * (1.0 / (1.0 + w.getActiveTaskCount()));
            return new WorkerScore(w.getId(), w.getUserId(), distanceKm,
                ↪ w.getActiveTaskCount(), score);
        })
        .sorted(Comparator.comparingDouble(WorkerScore::score).reversed())
        .collect(Collectors.toList());
}

public record WorkerScore(Long workerId, Long userId, double distanceKm, int
    ↪ activeTaskCount, double score) {}
}

```

Note on the “PageRank-style extension” mentioned in the pitch: if a judge asks how you’d extend the reporter-trust concept, you can describe (without necessarily implementing) a PageRank-like propagation where a citizen’s trust score rises when their reports get verified/resolved and falls when reports are marked invalid — an eigenvector-centrality-style update over a citizen→verified-report graph, conceptually the same centrality math already used in Section 11.3, just applied to a different graph. This is a strong “we thought about how to extend it” answer even as a described-not-built stretch.

12. Stretch Feature Specifications

12.1 Reporter Trust Score

Start simple: $\text{trustScore} = \text{verifiedReports} / \max(1, \text{totalReports})$, initialized to 1.0 for new users. Multiply a report's effective weight in the dedupe/priority pipeline by the reporting citizen's trustScore, so a low-trust account's reports still get recorded but contribute less to cluster_size. Describe the PageRank-style extension (Section 11.6 note) in Q&A if you don't have time to build it.

12.2 City Health Score

Per ward (a simple polygon or even a bounding-box grid cell for the demo area), aggregate open-issue counts by category into sub-scores, then combine:

```
roadScore      = 100 - min(100, openPotholeCount * 4)
lightingScore  = 100 - min(100, openStreetlightCount * 5)
waterScore     = 100 - min(100, openWaterCount * 6)
garbageScore   = 100 - min(100, openGarbageCount * 3)
safetyScore    = 100 - min(100, openSafetyCount * 8)
```

```
wardHealthScore = average(roadScore, lightingScore, waterScore, garbageScore, safetyScore)
```

Display as a ranked list or a Leaflet choropleth (color wards from red at low scores to green at high scores). The weights (4, 5, 6, 3, 8) are placeholders — tune them once you see how many open issues your seeded demo data actually has per category so the score spread looks meaningful on stage.

12.3 Predictive Alerts (Flood/Waterlogging Risk)

```
floodRisk = f(forecastRainfallMm, wardDrainageComplaintCount, wardWaterloggingComplaintCount)
```

A simple, defensible version: pull a rainfall forecast from Open-Meteo (no API key needed) for the demo area; if forecast rainfall exceeds a threshold (e.g., 40mm/24h) AND the ward has more than N open drainage/waterlogging complaints (from your own issue_clusters table filtered by issueType), display a banner: *"Heavy rainfall expected. Ward has 23 drainage complaints — Flood Risk: HIGH."* This is proactive rather than reactive and reuses data you already have — no new data pipeline required beyond one weather API call.

12.4 Before/After Resolution Verification

Send both the before_photo_url (from the original report) and after_photo_url (from the worker's resolution submission) to Gemini's vision endpoint with a prompt like: *"Compare these two images of the same location. Has the reported issue (a {issueType}) been visibly resolved? Respond with JSON: {"resolved": true|false, "confidence": 0-1}."* Store the result in resolutions.ai_confidence. **Implementation risk note:** this is the highest-risk stretch feature — vision comparison prompts can be inconsistent, and it depends on citizens' and workers' photos being framed similarly. Build and test this early if you plan to demo it live, or pre-record this specific interaction as a backup clip for the video.

12.5 SLA / Accountability Tracking

Define a per-category target resolution time (e.g., POTHOLE: 7 days, STREETLIGHT: 3 days, WATER: 2 days). On the dashboard, compute $\text{daysOpen} = \text{today} - \text{firstReportedAt}$ and flag

any cluster where `daysOpen > targetDays[issueType]` with a red “Overdue” badge. Trivial to compute, strong visual impact for the Impact/Viability criterion.

12.6 Citizen Reward System / Leaderboard

Award `reward_points` when: a citizen’s report is verified as valid (+10), a citizen’s report leads to a new cluster that gets resolved (+15), a citizen confirms a resolution is accurate (+5). Expose `GET /analytics/leaderboard` (top N citizens by `reward_points`) and a simple leaderboard screen in the citizen app.

12.7 AI Civic Copilot

A thin chat wrapper: forward the citizen’s natural-language question plus relevant context (their own report list, ward-level stats) to Gemini with a system prompt like *“You are a civic assistant. Answer only using the provided data. If you don’t know, say so.”* Build this only after everything else works — it’s a UI layer over data you already have, not new backend logic.

12.8 AI Voice Complaint Intake

Use the browser’s built-in Web Speech API (`SpeechRecognition`) for free, no-key voice-to-text on the citizen app; feed the resulting transcript into the same `description` field used by Step 5’s report form, so Step 13’s classification pipeline handles it identically to a typed description. Frame this explicitly as an accessibility feature for elderly or low-literacy users in your pitch.

12.9 Route Safety Score

A natural extension of Section 11.4: for a given home→destination query, run both fastest and safest Dijkstra passes, then compute:

$$\text{safetyScorePercent} = 100 \times (1 - \text{openFlaggedEdgesOnRoute} / \text{totalEdgesOnRoute})$$

Display alongside qualitative road-quality/streetlight-coverage labels derived from the same edge metadata already in your graph (Section 11.3) — no new data source required.

13. Testing Plan

Per Rule 16 of the Technical Rules, at minimum verify: main user flow, input validation, API connectivity, database operations, error handling, and responsiveness where applicable. Concretely, before submission run through this checklist:

- ☐ Submit a report as a citizen → appears correctly in the DB and on the dashboard.
- ☐ Submit two reports of the same category within 50m/3 days → confirm they merge into one cluster with `reportCount = 2`, not two separate clusters.
- ☐ Submit a report far away or of a different category → confirm it does **not** merge.
- ☐ Verify priority score changes sensibly as severity/cluster size/age change (spot-check the formula by hand on 2-3 examples).
- ☐ Request a route with `mode=fastest` and `mode=safest` between the same two points → confirm the paths differ once a flagged report exists on the fastest path's edges.
- ☐ Assign a worker manually, and (if built) via the suggestion endpoint → confirm `assigned_worker_id` and status update correctly.
- ☐ Mark a task resolved as a worker → confirm the citizen's tracking view reflects `RESOLVED`.
- ☐ Turn off/blank the Gemini API key → confirm the keyword fallback still classifies reports without crashing (Rule 11 Q&A readiness).
- ☐ Submit a report with missing required fields (no photo, no location) → confirm a clear validation error, not a 500.
- ☐ Load the citizen map and dashboard on a phone-width browser window → confirm layouts don't break (Rule/criteria: Design & UX, 15%).
- ☐ Restart the backend and confirm the road graph reloads from `road_graph.json` without needing to re-run the Python script.

14. Security & Deployment Notes

- **Never commit secrets.** `GEMINI_API_KEY`, `DB_PASSWORD`, `JWT_SECRET` all come from environment variables (see `.env.example` in Section 16), consistent with Technical Rules 9-10.
- **JWT auth** (Step 17) should use short-lived access tokens; for a hackathon, a single non-expiring demo token per role is acceptable but say so explicitly if asked — don't claim production-grade security you didn't build.
- **Rate-limit** the `/reports` and `/route` endpoints minimally (even an in-memory token bucket) if you have time — mention as a “future work” item otherwise.
- **Deployment (Rule 15):** deploy the Spring Boot backend + PostgreSQL to Render or Railway (both have a free tier sufficient for a demo), and each frontend app to Vercel or Netlify. Keep the deployed link live through evaluation, and if any screen requires login, provide demo credentials to judges as required by Rule 15.
- **CORS:** enable CORS on the backend for your deployed frontend origins explicitly (don't leave it wide open to `*` if you can avoid it — a judge asking about security is a very plausible Q&A moment given Rule 17 explicitly lists “security” as a topic they may probe).

15. Compliance Checklist (Mapped to the Technical Rules PDF)

Rule	Requirement	How this plan satisfies it
1	One public GitHub repo, organizers as collaborators	Section 6.1 repo layout; create the repo before Step 1 and invite organizers immediately.
2	Every member ≥ 2 meaningful commits	Section 18's task split assigns each member their own steps from Section 10.
3-4	Commit every 1-2 hrs, no last-minute dump	Section 10's steps are small; commit after each verified step, not at the end.
5	Branching for major features	Suggest: feature/citizen-app, feature/backend-core, feature/routing-graph, feature/dashboard, feature/worker-app, merged into main.
6	README.md with problem, solution, features, stack, architecture, APIs, DB, setup, run instructions	Template in Section 16.
7	Working prototype/MVP, demonstrable core features	Section 3.1's Must-Ship list = your MVP scope; don't pitch anything from Section 3.2 as done unless it's actually built and demoable.
8	Build & run instructions, reproducible	Section 16's README template includes exact setup/run commands.
9	Disclose APIs/libraries/frameworks used	Listed explicitly in Section 6 and the README template.
10	No secrets in source/GitHub; .env + .env.example	Section 16 gives the exact .env.example to commit (with placeholder values only).
11	Disclose significant AI-generated code/content in the PPT	Section 17 includes suggested disclosure wording for the pitch deck.
12	No plagiarism; identify original vs third-party components	Be explicit in the README about which parts are original algorithms (dedupe, priority scoring, worker assignment — all original to your team) vs third-party (OSM data, NetworkX centrality implementation, Gemini API, Leaflet).

Rule	Requirement	How this plan satisfies it
13	Version freeze at deadline	Stop pushing once the deadline hits; don't rely on late pushes being considered.
14	Final submission bundle	GitHub link + README + PPT + demo/deployment link + credentials — checklist in Section 16.
15	Deployment link, stay accessible, demo credentials	Section 14.
16	Testing before submission	Section 13's checklist.
17-18	Explain architecture/algorithms; each member explains their contribution	Because every algorithm here is a plain formula or textbook algorithm (Haversine, Dijkstra, Brandes' centrality) rather than an opaque model, every team member can genuinely explain their piece — rehearse the explanations in Section 17's Q&A prep.

16. README.md Template & .env.example

```
# CivicPulse – Intelligent Civic Issue Lifecycle & Risk-Aware Routing Platform

## Problem Statement
[Copy the SmartCity Tech problem statement summary from Section 2.1, then 2-3 sentences
on the specific pain point you're targeting for your demo area.]

## Proposed Solution
[2-3 paragraphs: the citizen -> AI -> department -> priority -> worker -> route ->
  ↪ resolution
pipeline, and the shared road graph that also powers fastest/safest routing. Reference
Section 1's executive summary.]

## Features
- Photo + GPS civic issue reporting with automatic duplicate/cluster detection
- Deterministic, explainable priority scoring (severity x cluster size x recency x
  ↪ criticality)
- Real road-graph-based routing with a fastest/safest toggle
- Authority dashboard: verify AI classification, prioritize, assign/reassign workers
- Worker app: prioritized task list, one-tap navigation, before/after resolution
  ↪ capture
- [List any Section 12 stretch features you actually completed]

## Tech Stack
- Backend: Java 17, Spring Boot 3, PostgreSQL 15
- Frontend: React 18, Vite, Leaflet.js, Tailwind CSS
- Offline graph preprocessing: Python 3, OSMnx, NetworkX (Brandes' betweenness
  ↪ centrality)
- AI: Google Gemini API (issue classification), with a keyword-based fallback
- Auth: JWT (Spring Security)

## System Architecture
[Paste/render the Mermaid diagram from Section 4.1]

## APIs
[Paste the endpoint table from Section 9]

## Database
[Paste the schema from Section 8, or a summary + link to schema.sql]

## Setup Instructions
### Prerequisites
- Java 17+, Maven, Node.js 18+, PostgreSQL 15, Python 3.10+

### Backend
``bash
cd backend
cp ../.env.example ../.env    # fill in real values
mvn spring-boot:run
```

Graph preprocessing (run once)

```
cd graph-preprocessing
pip install -r requirements.txt
python build_graph.py --place "<your demo area>" --out
  ↪ ../backend/src/main/resources/road_graph.json
```

Frontend

```
cd frontend-citizen    # repeat for frontend-dashboard, frontend-worker if separate apps
npm install
npm run dev
```

How to Run the Project

1. Start PostgreSQL and create the civicpulse database.
2. Run the graph preprocessing script once (see above).
3. Start the backend (mvn spring-boot:run), confirm it's up at <http://localhost:8080>.
4. Start the frontend(s) (npm run dev), open <http://localhost:5173>.
5. Register a CITIZEN, AUTHORITY, and WORKER account (or use seeded demo accounts — see below).

Third-Party APIs, Libraries & Datasets

- OpenStreetMap / Overpass API (via OSMnx) — road network data, no key required
- Google Gemini API — issue classification (optional; keyword fallback included)
- Leaflet.js + OpenStreetMap tiles — mapping, no key required
- Open-Meteo (if predictive alerts implemented) — weather forecast, no key required

AI-Assisted Development Disclosure

[State plainly which parts were AI-assisted (e.g., “boilerplate CRUD scaffolding and this implementation plan were AI-assisted; the dedupe, priority-scoring, and routing algorithms were designed and implemented by the team and every member can explain them line-by-line”).]

Demo Credentials

- Citizen: demo.citizen@civicpulse.test / [password]
- Authority: demo.authority@civicpulse.test / [password]
- Worker: demo.worker@civicpulse.test / [password]

Deployment

- Backend: [Render/Railway URL]
- Frontend: [Vercel/Netlify URL]

```
```bash
.env.example (commit this file; never commit the real .env)
DB_URL=jdbc:postgresql://localhost:5432/civicpulse
DB_USER=postgres
DB_PASSWORD=changeme
JWT_SECRET=replace-with-a-long-random-string
GEMINI_API_KEY=
```

## 17. Demo Script & Pitch Prep

### 17.1 Positioning (don't undersell it)

The rulebook's own pitch advice applies directly here: don't describe this as *"an AI complaint management system"* — that's ordinary and undersells the graph work. Pitch it as:

"An intelligent civic issue lifecycle platform that converts citizen observations into prioritized, location-aware tasks and coordinates their resolution across city authorities and field workers — built on a shared, risk-aware road graph that also tells citizens the safest way home."

The innovation is the **whole pipeline** (Citizen → AI → Department → Prioritization → Worker → Route → Resolution → Citizen) and the **shared graph**, not any single AI call.

### 17.2 5-minute demo video shot list (Rule: Final Submission #2, ≤5 minutes)

Time	Shot	Shows
0:00-0:30	Problem framing (voiceover over a slide or the rulebook's own SmartCity paragraph)	Why "report and pin" alone isn't enough
0:30-1:15	Citizen submits a pothole report (photo + GPS) live	Mandatory functional UI (Rule 7)
1:15-1:45	A second, nearby report submitted → dashboard shows it merged into the same cluster, report count went up	Duplicate detection (Section 11.1)
1:45-2:15	Dashboard: priority list, explain the formula briefly on-screen (Section 11.2)	Technical Implementation criterion
2:15-3:00	Route planner: toggle Fastest → Safest, visibly different route drawn, mention the flagged stretch it avoids	Innovation & Originality criterion — this is your best visual beat
3:00-3:30	Assign worker (manual or suggested), switch to Worker App, tap Navigate	Worker interface, end-to-end flow
3:30-4:00	Mark resolved, before/after photos, citizen sees "Resolved"	Closes the loop
4:00-4:30	Quick montage of any completed stretch features (heatmap / health score / predictive alert)	Impact & Viability
4:30-5:00	Closing line restating the positioning from 17.1 + tech stack card	Memorable close

### 17.3 Anticipated Q&A and how to answer

- *“Why not just use a vote count for priority?”* → Explain the formula (Section 11.2) and the worked example (severity × cluster size × recency × criticality).
- *“How does duplicate detection work, isn’t that expensive?”* → Haversine + time window,  $O(k)$  per report against currently-open clusters of that type (Section 11.1); mention the PostGIS/geohash scaling path.
- *“What if the AI classification is wrong?”* → Authority can verify/correct it (Step 14); Gemini only classifies, your backend decides everything downstream (Section 11.5).
- *“What if the Gemini API is down during judging?”* → Keyword fallback (Step 13) keeps the flow working; demo this explicitly if asked.
- *“Is this real data?”* → Yes for the road graph (real OSM extract for your demo area, Section 11.3); be upfront about which parts (e.g., seeded workers/citizens) are demo data.
- *“How would this scale to a whole city?”* → Spatial indexing for dedupe (PostGIS ST\_DWithin), batching centrality recomputation nightly rather than per-request, sharding the graph by ward.
- *“What’s actually novel here vs. a normal complaint app?”* → The shared graph powering both prioritization and routing, plus duplicate clustering — walk through Section 4.3’s “why one shared graph powers two products” explanation.

## 18. Team Task Split & Hour-by-Hour Timeline (for a 3-5 member team)

This assumes a 24–36 hour hackathon window; scale proportionally to your actual duration. Adjust member count to your real team size (rulebook requires 3–5 members).

### 18.1 Suggested role split

Member	Owns	Steps from Section 10
Member A (backend/algorithms lead)	Data model, dedupe, priority scoring, worker assignment	Steps 1, 2, 3, 8, 9, 15, 17
Member B (graph/routing specialist — good fit for a DSA/competitive-programming background)	Offline graph build, Dijkstra/A*, routing endpoint	Steps 10, 11, 12
Member C (frontend — citizen-facing)	Citizen report form, map, route planner UI	Steps 5, 6, 12 (UI half)
Member D (frontend — dashboard/worker)	Authority dashboard, worker app	Steps 4, 7, 14, 16
Member E (AI integration + polish, or absorbed into A/C if team of 3–4)	Gemini classification, stretch features, demo video/deck	Steps 13, 18–20, Section 17

Every member touches at least 2 distinct steps with real commits, satisfying Rule 2.

### 18.2 Rough timeline (24-hour example)

Hours	Focus
0–2	Repo setup (Step 1), agree on data model (Step 2), pull OSM data for the demo area and sanity-check it (start of Step 10)
2–6	Backend core (Steps 3, 4), citizen report form (Step 5), dashboard skeleton (Step 7) in parallel
6–10	Dedupe + priority scoring (Steps 8, 9), citizen map (Step 6), graph loading + routing (Steps 10, 11)
10–14	Route planner UI + fastest/safest toggle (Step 12) — get this working early, it's your best demo beat
14–18	Authority verify/assign (Step 14), worker app (Step 16), auth (Step 17)
18–21	Stretch features from Section 3.2 top-down, as time allows (Steps 13, 15, 18–20)
21–23	Full run-through testing (Section 13 checklist), fix bugs, seed clean demo data
23–24	Record the 5-minute demo video (Section 17.2), finalize README (Section 16), freeze the repo

**Golden rule:** stop adding stretch features at least 2–3 hours before the deadline and spend that time on testing, demo data quality, and the video/pitch — a polished Must-Ship demo beats a half-working stretch feature every time under this rubric (Design & UX is 15%, and a broken live demo costs far more than a missing feature ever would).

## 19. Appendix

### 19.1 Seed data script (for a clean, reliable demo)

```
-- backend/src/main/resources/seed-demo-data.sql (run manually against your dev DB
↳ before demoing)
INSERT INTO users (name, email, password_hash, role) VALUES
 ('Demo Citizen', 'demo.citizen@civicpulse.test', '$2a$10$replace_with_bcrypt_hash',
 ↳ 'CITIZEN'),
 ('Demo Authority', 'demo.authority@civicpulse.test',
 ↳ '$2a$10$replace_with_bcrypt_hash', 'AUTHORITY'),
 ('Raj Kumar', 'raj@civicpulse.test', '$2a$10$replace_with_bcrypt_hash', 'WORKER');

INSERT INTO workers (user_id, department, current_lat, current_lng, active_task_count)
↳ VALUES
 ((SELECT id FROM users WHERE email = 'raj@civicpulse.test'), 'ROAD', 23.1091,
 ↳ 72.5343, 1);

-- Seed 2-3 clusters at real coordinates within your chosen OSM bounding box so the
-- fastest/safest route toggle has something visible to react to (Section 11.4's demo
↳ tip).
```

### 19.2 Glossary

- **Cluster (Issue Cluster):** one real-world civic issue, formed by merging one or more individual citizen reports that reference the same problem (Section 11.1).
- **Centrality (betweenness):** a graph metric describing how often a road segment lies on the shortest path between other points — used here as a proxy for “structurally important road” (Section 11.3).
- **Edge cost function:** the formula Dijkstra/A\* minimizes when finding a path; swapping this one function between “fastest” and “safest” is the entire trick behind the routing feature (Section 11.4).
- **Dedupe radius / time window:** the distance (default 50m) and time (default 3 days) thresholds used to decide whether a new report belongs to an existing cluster (Section 11.1).

### 19.3 Full repository layout (recap of Section 6.1)

```
civicpulse/
├── README.md
├── .env.example
├── .gitignore
├── backend/
├── graph-preprocessing/
├── frontend-citizen/
├── frontend-dashboard/
├── frontend-worker/
└── docs/
```

---

*End of document. Feed Section 10’s steps to your AI assistant one at a time, verify each before moving on, and keep this document open as the shared reference for architecture (Section 4), data model (Section 8), API contract (Section 9), and algorithm details (Section 11) whenever a step needs more context than fits in its own block.*